

PHYSICS 2024

(Science Pre-Engineering & Pre-Medical Groups)

TIME: 3 Hours

(85 Marks)

SECTION 'A' (Multiple Choice Questions) (17)

NOTE: i) This section consists of 17 part questions and all are to be answered. Each question carries 1 mark.
 ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.
 iii) The use of scientific calculator is allowed. All notations are used in their usual meanings

1. Select the correct answer for each from the given options:

i) A point charge $q_1 = 2\mu\text{C}$ is located at a distance 'd' away from another point charge $q_2 = 6\mu\text{C}$. The ratio $\frac{F_{12}}{F_{21}}$ will be:

* 1 ✓

* 3

* $\frac{1}{2}$ * $\frac{1}{3}$

ii) The work done by a centripetal force is always:

* Maximum

* Half of maximum value

* Quarter of maximum value

* Zero ✓

iii) If the cross-sectional area of a pipe decreases, then the velocity of fluid:

* Increases ✓

* Decreases

* Depends on the fluid density

* Remains the same

iv) The process of superimposing signal frequency on carrier wave is:

* Transmission

* Detection

* Reception

* Modulation ✓

v) A heat sensitive device whose resistivity changes with the change in temperature is called:

* Conductor

* Thermistor ✓

* Resistor

* Thermometer

vi) A ray of light enters obliquely from air into water then these two quantities of the following change:

(a) wavelength

(b) frequency

(c) speed

(d) direction:

* (a), (c) and (d) ✓

* (a) and (b)

* (b) and (d)

* (b) and (c)

vii) The dot product and cross product of two vectors are equal at angle:

* 65° * 45° ✓* 76° * 30°

viii) The rate of change of linear momentum is:

- * Force * Impulse✓ * Torque * Power

ix) One Degree (1°) is equal to:

- * π radians * $(180/\pi)$ radians
 * $(\pi/180)$ radians✓ * 2π radians

x) Dimensions of kinetic energy are the same as those of:

- * Force * Work✓ * Momentum * Power

xi) A completely submerged object always displaces fluid equal to its:

- * Weight✓ * Volume * Density * Mass

xii) In Young's double slit experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe spacing will be:

- * 4 times✓ * $\frac{1}{4}$ times * doubled * unchanged

xiii) A wire of uniform area of cross-section 'A' length 'L' and resistance 'R' is cut into two equal parts. The resistivity of each part.

- * Becomes zero * Is halved
 * Is doubled * Remains same✓

xiv) A child is swinging on a swing in sitting position. If he stands up, then the time period of the swing will:

- * Increase * Decrease✓
 * Remains the same * First increase and then decrease

xv) The charging of a capacitor through a resistance follows:

- * Linear law * Exponential law✓
 * Inverse square law * Power law

xvi) The physical quantity which remains constant during simple harmonic motion is:

- * Velocity * Acceleration
 * Energy✓ * Momentum

xvii) If V_a , V_h and V_m are the speed of sound in air, hydrogen and metal at same temperature, then:

- * $V_a > V_h > V_m$ * $V_m > V_h > V_a$ ✓
 * $V_h > V_m > V_a$ * $V_m > V_a > V_h$

SECTION 'B' (Short-Answer Questions) (36)

NOTE: Answer any NINE part questions from this section. All questions carry equal marks. Draw diagrams where necessary. Use of scientific calculator is allowed.

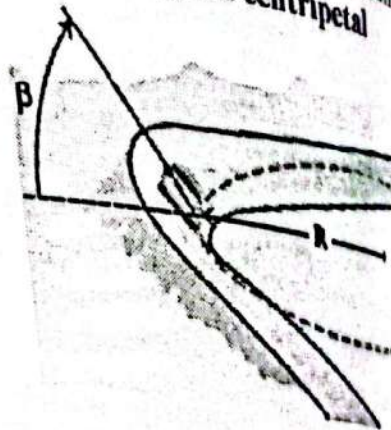
2. i) With the help of diagram show the forces acting on a car moving on a banked curve and derive the mathematical expression for banking.

$$\theta = \tan^{-1} \left(\frac{v^2}{rg} \right)$$

ANSWER:

FORCES ACTING ON BANKED CURVE

A banked curve is a curve that has its surface at angle with respect to ground on which the curve is positioned. The reason for banking curve to decrease the moving object depends on the force of friction. On a curve that is not banked, a car traveling along that curve will experience a static friction that will point towards the center of the circular path restricted by the moving car. This frictional force will be responsible for creating centripetal acceleration, which in turn will allow the car to move along the curve. On a banked curve however, the normal force acting on an object such as a car; will act at an angle with the horizontal, and thus create a component normal force that acts along the x axis. This component normal force will now be responsible for creating the centripetal acceleration required to move the car along the curve. Therefore, for every single angle, there exists a velocity for which no friction is required at all to move the object along the curve. This means that the car will be able to turn even under the most slippery conditions (ice or water).



DERIVATION

Consider a car on a frictionless banked curve. If the angle θ is ideal speed and radius r , then the net external force equals the necessary centripetal force. The only two external forces acting on the car are weight and the normal force of the road N . (A frictionless surface exerts a force perpendicular to the surface that is, a normal force.) These two forces must be added to give a net external force that is horizontal

toward the centre of curvature and has magnitude $\frac{mv^2}{r}$. As it is a centripetal force and is horizontal, we use a coordinate system with vertical and horizontal axes. Only the normal force has a horizontal component. This component must equal the centripetal force, that is,

$$N \sin \theta = \frac{mv^2}{r} \quad \text{--- (i)}$$

Because the car does not leave the surface of the road, the net vertical force must be zero, meaning that the vertical components of the two forces

forces must be equal in magnitude and opposite in direction. We note that the vertical component of the normal force is $N \cos \theta$, and the only other vertical force is the car's weight. These must be equal in magnitude; thus,

$$N \cos \theta = mg \quad \text{--- (ii)}$$

Now dividing equation (i) by equation (ii), we get

$$\begin{aligned} \frac{N \sin \theta}{N \cos \theta} &= \frac{\frac{mv^2}{r}}{mg} \\ \frac{\sin \theta}{\cos \theta} &= \frac{v^2}{rg} \\ \tan \theta &= \frac{v^2}{rg} \\ \theta &= \tan^{-1} \left(\frac{v^2}{rg} \right) \end{aligned}$$

This expression can be understood by considering how θ depends on v and r . A large θ is obtained for a large v and a small r .

ii) How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other is equal to the magnitude of the weight of a proton on earth's surface. ($m = 1.67 \times 10^{-27} \text{ kg}$, $q = 1.6 \times 10^{-19} \text{ C}$)

ANSWER:

DATA:

Mass of Proton = $M_p = 1.67 \times 10^{-27} \text{ kg}$

Charge of proton = $q_1 = q_2 = q = 1.6 \times 10^{-19} \text{ C}$

Magnitude of Force = $F = W = mg = (1.67 \times 10^{-27})(9.8)$
 $= 1.636 \times 10^{-26} \text{ N}$

Value of permittivity for vacuum = ϵ_0
 $= 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$

Distance between Protons = $r = ?$

SOLUTION:

$$\therefore \bar{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$1.636 \times 10^{-26} = \frac{1}{4 \left(\frac{22}{7} \right) (8.854 \times 10^{-12})} \frac{(1.6 \times 10^{-19})^2}{r^2}$$

$$r^2 = \frac{2.56 \times 10^{-38}}{1.112 \times 10^{-10}} \times \frac{1}{1.636 \times 10^{-26}}$$

$$r^2 = \frac{2.56 \times 10^{-38}}{1.819 \times 10^{-36}}$$

$$r^2 = 0.0140$$

Taking Square root on both sides

$$r = 0.118 \text{ m}$$

iii) State Kirchhoff's first and second law. Draw diagram for both the

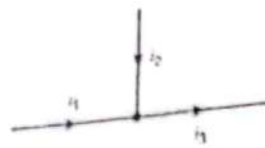
ANSWER:

KIRCHHOFF'S FIRST LAW

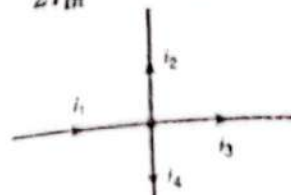
Kirchhoff's first law (current Law) states that "the current that flows into a junction-any electrical connection-must equal the current that flows out of the same junction".

Mathematically

$$\Sigma I_{in} = \Sigma I_{out}$$



(a)



(b)



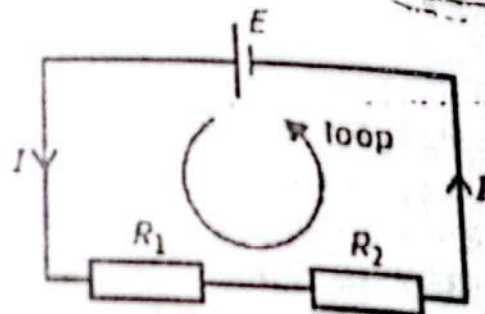
(c)

KIRCHHOFF'S SECOND LAW

The Kirchhoff's second law (voltage law) states that "the sum of electromotive forces in a loop equals the sum of potential drops in the loop".

Mathematically

$$\Sigma E = \Sigma V$$



iv) A mortar shell is fired at a ground level target 500m distance with initial velocity of 90 ms^{-1} . What is the launch angle?

ANSWER:

DATA:

Range = $R = 500 \text{ m}$

Initial Velocity = $V_o = 90 \text{ ms}^{-1}$

Acceleration due to gravity = $g = 9.8 \text{ ms}^{-2}$

launch angle = $\theta = ?$

SOLUTION:

$$\therefore R = \frac{V_o^2 \sin 2\theta}{g}$$

$$\therefore R \times \frac{g}{V_o^2} = \sin 2\theta$$

$$\sin 2\theta = 500 \times \frac{9.8}{(90)^2}$$

$$\sin 2\theta = \frac{4900}{8100}$$

$$\sin 2\theta = \frac{4900}{8100}$$

$$2\theta = \sin^{-1}(0.604)$$

$$\theta = \frac{37.2244}{2}$$

$$\theta = 18.61$$

$$\theta' = 90^\circ - \theta$$

$$\theta' = 90^\circ - 18.61^\circ$$

$$\theta' = 71.39^\circ \approx 71.4^\circ$$

$$\theta' = 71.4^\circ$$

1) How energy is stored in a capacitor? Derive mathematical expression for the energy stored in capacitor.

ANSWER:

ENERGY IS STORED IN A CAPACITOR

The capacitance of capacitors refers to the amount of accumulated electrical charge stored on the plates. The amount of energy stored in a capacitor depends on how much charge is stored on the plates.

Charging a capacitor always involves some energy expenditure by the charging agency. This energy is stored in the electrostatic field set up in the dielectric medium. On discharging the capacitor, the field collapses and the stored energy is released.

DERIVATION:

Consider a capacitor connected to source of potential difference V .

Initially, when the capacitor is uncharged, the potential difference between the plates is zero. Finally when charge $+Q$ and $-Q$ deposited on the plates, the potential difference between the plates becomes V .

$$\therefore V = \frac{E}{Q}$$

The average voltage on the capacitor during the charging process is $V/2$.

$$\frac{V}{2} = \frac{E}{Q}$$

$$E = \frac{1}{2} VQ$$

$$\therefore Q = CV$$

$$E = \frac{1}{2} V(CV)$$

$$E = \frac{1}{2} CV^2$$

vi) Define escape velocity. Derive the relation $v_{es} = \sqrt{2gR}$ for escape velocity on earth's surface.

ANSWER:

ESCAPE VELOCITY

Escape velocity on earth or any other planet is defined as "the minimum velocity with which the body has to be projected vertically upwards from the surface of the earth or any other planet so that it just crosses the gravitational field of earth or of that planet and never return on its own."

DERIVATION:

Work is done at the cost of kinetic energy given to the body at the surface of earth. If V_{es} is the escape velocity of the body. M and R are the mass and Radius of the earth respectively the body will escape out of the gravitational field.

$$\frac{1}{2} m(V_{es})^2 = \frac{GMm}{R}$$

$$(V_{es})^2 = \frac{2GM}{R}$$

On taking square roots on both sides

$$V_{es} = \sqrt{\frac{2GM}{R}} \quad \text{--- (i)}$$

$$\because g = \frac{GM}{R^2} \Rightarrow \frac{GM}{R} = gR$$

On substituting the value of $\frac{GM}{R}$ in equation (i), we get

$$V_{es} = \sqrt{2gR} \text{ Derived}$$

vii) The small piston of a hydraulic press has an area of 10cm^2 . If the force is 50N what must the area of the large piston to exert a pressure of 4800N .

ANSWER:

DATA:

Area of small piston = $A_1 = 10.0\text{cm}^2$

Force applied on small piston = $F_1 = 50.0\text{N}$

Force applied on Large piston = $F_2 = 4800\text{N}$

Area of Large piston = $A_2 = ?$

SOLUTION:

$$\because \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$A_2 = \frac{F_2}{F_1} \times A_1$$

$$A_2 = \frac{4800N}{50.0N} \times 10.0cm^2$$

$$A_2 = 960cm^2$$

(iii) A block is kept on a horizontal table. The table is undergoing simple harmonic motion of frequency 3Hz in a horizontal plane. The coefficient of static friction between the block and the table is 0.72. Find the maximum amplitude of the table at which the block does not slip on this table.

ANSWER:

DATA:

Frequency = $f = 3Hz$

Coefficient of static friction = $\mu = 0.72$

Amplitude = $x_0 = ?$

Acceleration due to gravity = $g = 9.8ms^{-1}$

SOLUTION:

$$\therefore a = \omega^2 x_0 \text{ --- (i)}$$

Maximum force of static friction is given as

$$\therefore F = \mu mg$$

In case that the body does not slip, therefore

$$ma = \mu mg$$

$$a = \mu g$$

On substituting the value of "a" in equation (i), we get

$$\mu g = \omega^2 x_0$$

$$x_0 = \frac{\mu g}{\omega^2}$$

$$\therefore \omega = 2\pi f$$

$$x_0 = \frac{\mu g}{(2\pi f)^2}$$

$$x_0 = \frac{\mu g}{4\pi^2 f^2}$$

$$x_0 = \frac{(0.72)(9.8)}{4\left(\frac{22}{7}\right)^2 (3)^2}$$

$$x_0 = \frac{7.056}{7.056}$$

$$x_0 = 1.0$$

$$x_0 = 0.01985m$$

OR

$$x_0 = 1.985cm$$

(iv) Derive Bragg's equation for X-ray diffraction.

ANSWER:

DERIVATION:

Constructive interference occurs if the path difference between X-rays reflecting from an adjacent pair of planes is an integral multiple of the

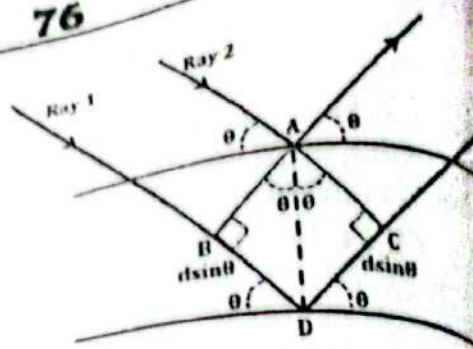
wavelength.

Figure shows that the path difference is $2d\sin\theta$, where d is the distance between the planes and θ is the angle that the incident and reflected beams make with the plane (not with the normal). Then, constructive interference occurs at angles given by Bragg's law:

$$p \cdot d = BD + CD = d\sin\theta + d\sin\theta = 2d\sin\theta$$

$$2d\sin\theta = m\lambda \text{ Derived}$$

$$(m = \pm 1, \pm 2, \pm 3, \dots)$$



x) Calculate the viscous drag on a drop of oil of 1mm radius falling air at its terminal velocity. (viscosity of air = 1.8×10^{-5} pass den = $850 \text{ g}^3/\text{m}^4$).

ANSWER:

DATA:

Radius of oil drop = $r = 0.1 \text{ mm} = \frac{0.1}{1000} = 1 \times 10^{-4}$

Viscosity of air = $\eta = 1.8 \times 10^{-5} \text{ Pa.s}$

Density of oil = $\rho_{\text{oil}} = 850 \text{ kg/m}^3$

Density of air = $\sigma = 1 \text{ kg/m}^3$

Acceleration due to gravity = $g = 9.8 \text{ ms}^{-2}$

Viscous drag = $F = ?$

SOLUTION:

Since the terminal velocity of a drop is

$$\therefore V_t = \frac{2gr^2}{9\eta} (\rho - \sigma)$$

$$V_t = \frac{(19.8)(1 \times 10^{-4})^2}{9(1.8 \times 10^{-5})} (850 - 1)$$

$$V_t = \frac{(19.6)(1 \times 10^{-8})(849)}{1.62 \times 10^{-4}} = \frac{1.664 \times 10^{-4}}{1.62 \times 10^{-4}}$$

$$V_t = 1.0271$$

$$\therefore F = 6\pi\eta r v$$

$$F = 6\pi(1.8 \times 10^{-5})(1 \times 10^{-4})(1.0271)$$

$$F = 3.485 \times 10^{-8} \text{ N}$$

xi) A 20 meter long wire has a cross sectional area of 1 mm^2 and resistance of . Calculate the conductivity of the wire.

ANSWER:

DATA:DC resistance = $R = 5 \text{ ohms}$ Cable length = $L = 20 \text{ m}$ Cross-sectional area of the conductor = $A = 1 \text{ mm}^2$
 $= 1 \times 10^{-6} \text{ m}^2$ Conductivity = $\sigma = ?$ **SOLUTION:**

$$\begin{aligned} \therefore \sigma &= \frac{L}{RA} \\ \sigma &= \frac{20 \text{ m}}{(5)(1 \times 10^{-6} \text{ m}^2)} \\ \sigma &= \frac{20}{5 \times 10^{-6}} \\ \sigma &= 4000000 \text{ S/M} \end{aligned}$$

OR

$$\sigma = 4 \text{ MSm}^{-1}$$

Q) State the law of conservation of linear momentum

By using the law, calculate the speed of recoil of the gun, when a 100g bullet is fired from a 10kg gun with a speed of 1000m/s.

ANSWER:**CONSERVATION OF LINEAR MOMENTUM**

The law of conservation of linear momentum states that "if no external forces act on the system of two colliding objects, then the vector sum of the linear momentum of each body remains constant and is not affected by their mutual interaction".

DATA:Mass of bullet = $m = 100 \text{ g} = 0.1 \text{ kg}$ Mass of gun = $M = 10 \text{ kg}$ Speed of bullet = $V_1 = 1000 \text{ m/s}$ Initial velocity of Bullet = $U_1 = 0$ Initial velocity of Gun = $U_2 = 0$ Speed of recoil = $V_2 = ?$ **SOLUTION:**

Applying to law of conservation of momentum

$$\begin{aligned} mU_1 + MU_2 &= mV_1 + MV_2 \\ 0 + 0 &= (0.1)(1000) + 10V_2 \\ 10V_2 &= -100 \end{aligned}$$

$$V_2 = -\frac{100}{10} = -10 \text{ m/s}$$

Speed of recoil of gun is 10 m/s and negative sign shows opposite direction of motion.

Q) Prove that following equations are dimensionally correct.

$$* T = 2\pi \sqrt{\frac{l}{g}}$$

$$* x = m \frac{\lambda}{2}$$

ANSWER:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

PROOF:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

On substituting the dimensions of various quantities in the above we get

$$T = 2\pi \left(\frac{L}{LT^{-2}} \right)^{\frac{1}{2}}$$

In the above equation 2π is dimensionless constant hence therefore

$$T = (L^{1-1} T^{-2})^{\frac{1}{2}} = (L^0 T^{-2})^{\frac{1}{2}}$$

$$T = T$$

Each term on both sides of the above equation has same dimension. Proved the given expression is dimensionally correct.

$$x = m \frac{\lambda}{2}$$

PROOF:

$$x = m \frac{\lambda}{2}$$

On substituting the dimensions of various quantities in the above we get

$$L = m \left(\frac{L}{2} \right)$$

In the above equation " m " and " 2 " is dimensionless constant therefore

$$L = L$$

Hence Proved the given expression is dimensionally correct.

xiv) Write Newton's formula for the speed of sound in air. What did Laplace made in it? Also give the corrected formula.

ANSWER:

Newton's formula for speed of sound in air is:

$$\text{Newton's formula} = v = \sqrt{\frac{P}{\rho}}$$

Newton assumed that the propagation of sound waves is an adiabatic process. But According to Laplace when sound waves travel the process is isothermal.

there is compression and rarefaction in the particles of the medium. Where there is compression, particles come near to each other and the temperature rises. At rarefaction, particles go apart and there is a fall in temperature. Therefore, the temperature does not remain constant when sound waves travel through air or any other gas. As sound waves travel through the air, the changes in air pressure, volume and temperature take place rapidly. Therefore this process is adiabatic, not isothermal, so, Boyle's law is not applicable. The total quantity of heat in the system as a whole remains constant. It neither gains nor loses any heat through the surroundings.

CORRECTED FORMULA

$$\gamma P = \sqrt{\frac{\Delta P}{\rho}}$$

SECTION "C" (Detailed Answer Questions)(32)

NOTE: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

3. a) What is Bernoulli's principle? Derive Bernoulli's equation for steady, incompressible, non-viscous and irrotational flow of fluid.

ANSWER:**BERNOULLI'S PRINCIPLE**

Bernoulli's principle states that "The velocity of a fluid is high, the pressure is low, and the velocity is low, the pressure is high".

OR

Bernoulli's principle states that "The total mechanical energy of the moving fluid comprising the gravitational potential energy of elevation, the energy associated with the fluid pressure and the kinetic energy of the fluid motion, remains constant".

DERIVATION:

Consider a pipe whose diameter and elevation change as a fluid passes through it. Consider a small mass of the fluid with density ρ that flows from point 1 to 2 as shown in figure. The work done by a force F on the fluid to displace it by an infinitesimal distance Δx is given by,

$$W = F\Delta x$$

Therefore, at points 1 and 2, the work done are,

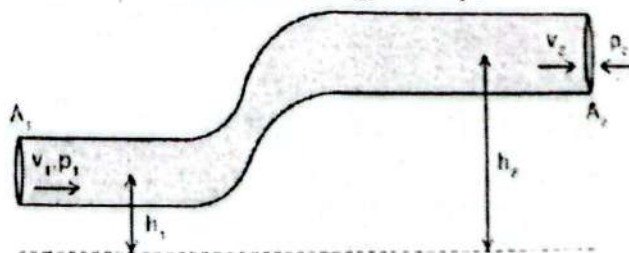
$$\Delta W_1 = F_1 \Delta x_1$$

$$\Delta W_2 = F_2 \Delta x_2$$

Total work done when the fluid moves 1 to 2 is,

$$\Delta W = \Delta W_1 - \Delta W_2$$

$$\therefore W = F \cdot x$$



Or,

$$\Delta W = F_1 \Delta x_1 - F_2 \Delta x_2$$

$$\Delta W = (p_1 A_1) \Delta x_1 - (p_2 A_2) \Delta x_2$$

$$\Delta W = p_1 A_1 \Delta x_1 - p_2 A_2 \Delta x_2$$

$$\Delta W = p_1 (A_1 \Delta x_1) - p_2 (A_2 \Delta x_2)$$

$$= p_1 \Delta V - p_2 \Delta V$$

$$= (p_1 - p_2) \Delta V$$

Now, when the fluid moves from point 1 to 2, there is a change of kinetic energy.

$$\Delta K.E = \frac{1}{2} m_2 v_2^2 - \frac{1}{2} m_1 v_1^2$$

$$= \frac{1}{2} (\rho \Delta V) v_2^2 - \frac{1}{2} (\rho \Delta V) v_1^2$$

$$\Delta K.E = \frac{1}{2} \rho \Delta V (v_2^2 - v_1^2)$$

Similarly, the change in potential energy is given by,

$$\Delta U = mgh_2 - mgh_1$$

$$\Delta U = mg(h_2 - h_1)$$

$$= \rho \Delta V g(h_2 - h_1)$$

The work done in moving the fluid is the sum of the change in kinetic and potential energies.

$$\Delta W = \Delta K.E + \Delta U$$

$$(p_1 - p_2) \Delta V = \left\{ \frac{1}{2} \rho \Delta V (v_2^2 - v_1^2) \right\} + \{ \rho \Delta V g(h_2 - h_1) \}$$

$$(p_1 - p_2) \Delta V = \frac{1}{2} \rho \Delta V (v_2^2 - v_1^2) + \rho \Delta V g(h_2 - h_1)$$

$$(p_1 - p_2) \Delta V = \Delta V \left\{ \frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g(h_2 - h_1) \right\}$$

$$p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g(h_2 - h_1)$$

$$p_1 - p_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 + \rho gh_2 - \rho gh_1$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho gh_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho gh_2$$

$$p + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

b) What is an electric dipole? Derive the expression for electric field intensity due to electric dipole at a point which is at a perpendicular distance y from the centre of the dipole.

ANSWER:

ELECTRIC DIPOLE:

An electric dipole is a simple system in electromagnetism consisting of two opposite electric charges of equal magnitude, separated by a small distance. The charges create an electric field that has a distinct pattern, with field lines oriented along the axis of the dipole.

ELECTRIC FIELD DUE TO AN ELECTRIC DIPOLE

DERIVATION:

A pair of equal and opposite point charges separated by a small distance form an electric dipole. To calculate the electric field of the dipole at point C which is at a distance "y" from the center of dipole. Consider two charges "-q" and "+q" placed at a small distance "d" from each other as shown in figure. The charge "+q" sets up field E_+ and the charge "-q" produces the field E_- .

Resolving the field vectors into the components. From figure, it is clear that the vertical components " $E_+ \cos \theta$ " and " $E_- \cos \theta$ " cancel each other. Therefore, the net electric field is due to the vector sum of the horizontal components.

$$\vec{E} = E_+ \cos \theta + E_- \cos \theta$$

The magnitude of both electric fields is the same. i.e.

$$E_+ = E_- = E$$

$$\vec{E} = E \cos \theta + E \cos \theta$$

$$\vec{E} = 2E \cos \theta$$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\vec{E} = 2 \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \cos \theta \quad \text{--- (i)}$$

From figure, In triangle ADC,

$$\cos \theta = \frac{d/2}{r} = \frac{d}{2r}$$

Putting the value of $\cos \theta$ in equation (i), we

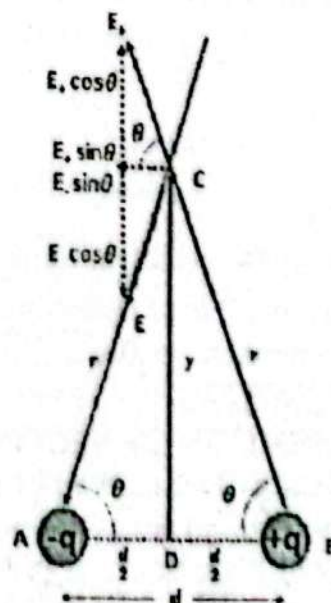
$$\vec{E} = 2 \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \left(\frac{d}{2r} \right)$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^3} \quad \text{--- (ii)}$$

From figure, In triangle ADC,

Pythagoras theorem

$$r^2 = y^2 + \left(\frac{d}{2} \right)^2$$



$$r = \sqrt{y^2 + \frac{d^2}{4}}$$

On substituting the value r in equation (ii), we get

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{1}{dq} \left(\sqrt{y^2 + \frac{d^2}{4}} \right)^{\frac{3}{2}}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{1}{dq} \frac{d^2}{\left(y^2 + \frac{d^2}{4}\right)^{\frac{3}{2}}}$$

Since the product of the "q" and "d" is known as the electric dipole moment "P" of the dipole, i.e. $dq = P$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{P}{\left(y^2 + \frac{d^2}{4}\right)^{\frac{3}{2}}}$$

The electric dipole moment is a vector quantity and it is always from the negative to the positive charge.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{P}{\left\{y^2 \left(1 + \frac{d^2}{4y^2}\right)\right\}^{\frac{3}{2}}}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3 \left(1 + \frac{d^2}{4y^2}\right)^{\frac{3}{2}}}$$

As d is very small as compare to y ($d \ll y$), therefore approx field intensity of the dipole can be calculated by neglecting the

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3 (1)^{\frac{3}{2}}}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3} \quad \text{Derived}$$

4. a) Describe the addition of vectors by Head to Tail method vectors $\vec{P}$ and $\vec{Q}$ making angles θ_1 and θ_2 with the horizontal component method.

ANSWER:

ADDITION OF VECTORS BY HEAD TO TAIL METHOD

Head to Tail method or graphical method is one of the easiest used to find the resultant vector of two or more than two vectors.

Consider two vectors $\vec{A}$ and $\vec{B}$, acting in the directions as shown in figure.

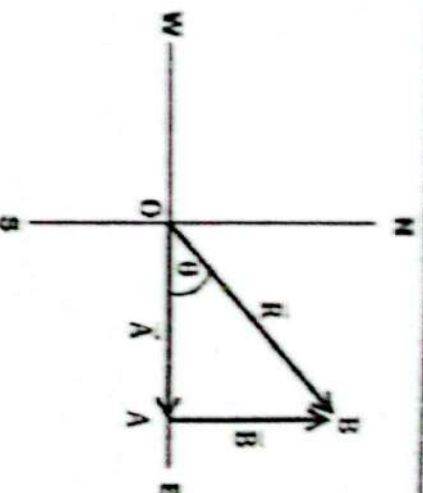
Head to tail rule is a method of vector addition in which tail of second vector is connected by head of first vector. All vectors are connected in this way.

Finally, from tail of first vector to the head of last vector we will draw a vector called resultant vector.

Mathematically for resultant vector ' $\vec{R}$ '

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{R} = \vec{A} + \vec{B}$$



ADDITION OF VECTORS BY RECTANGULAR COMPONENTS

METHOD:

Consider two vectors $\vec{P}$ and $\vec{Q}$ making angles θ_1 and θ_2 with +ve x-axis respectively.

Resolve vector $\vec{P}$ into two rectangular components $\vec{P}_x$ and $\vec{P}_y$ as shown in figure

Magnitude of these components are:

$$P_x = P \cos \theta_1$$

And

$$P_y = P \sin \theta_1$$

Resolve vector $\vec{Q}$ into two rectangular component $\vec{Q}_x$ and $\vec{Q}_y$ as shown in figure

Magnitude of these components are:

$$Q_x = Q \cos \theta_2$$

And

$$Q_y = Q \sin \theta_2$$

Now move vector $\vec{Q}$ parallel to itself so that its initial point (tail) lies on the terminal point (head) of vector $\vec{P}$ as shown in the figure

Representative lines of $\vec{P}$ and $\vec{Q}$ are OA and OB respectively. Join O and B which is equal to resultant vector $\vec{P}$ and $\vec{Q}$

Resultant vector along x - axis can be determined as:

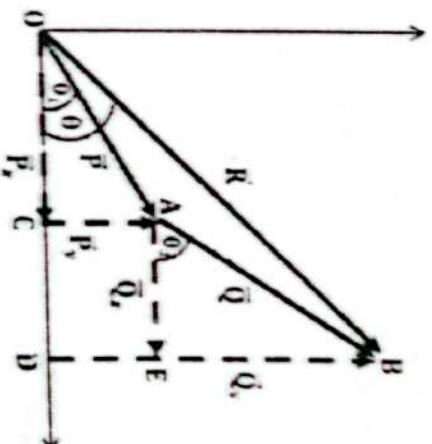
$$\vec{OD} = \vec{OC} + \vec{CD}$$

$$\therefore \vec{CD} = \vec{AE}$$

$$\vec{OD} = \vec{OC} + \vec{AE}$$

$$\vec{R}_x = \vec{P}_x + \vec{Q}_x$$

$$\vec{R}_x = P \cos \theta_1 + Q \cos \theta_2$$



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Resultant vector along y-axis can be determined as:

$$\overrightarrow{OD} = \overrightarrow{DE} + \overrightarrow{EB}$$

$$R_y = P_y + Q_y$$

$$R_y = P \sin \theta_1 + Q \sin \theta_2$$

Now we will determine the magnitude of resultant vector by using Pythagoras' theorem.

$$\text{Hypotenous}^2 = \text{Base}^2 + \text{Perpendicular}^2$$

$$|\overrightarrow{OB}|^2 = |\overrightarrow{OD}|^2 + |\overrightarrow{DB}|^2$$

$$R^2 = R_x^2 + R_y^2$$

$$R = \sqrt{R_x^2 + R_y^2}$$

$$R = \sqrt{(P \cos \theta_1 + Q \cos \theta_2)^2 + (P \sin \theta_1 + Q \sin \theta_2)^2}$$

Finally the direction of resultant vector will be determined.

Again in the right angled triangle $\angle DOB$:

$$\tan \angle DOB = \frac{DB}{OD}$$

$$\tan \angle DOB = \tan \theta = \frac{R_y}{R_x}$$

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$$

Where θ is the angle that the resultant vector makes with the positive axis. In this way we can add a number of vectors in a very easy manner. This method is known as addition of vectors by rectangular component method.

b) Name the acceleration due to the change in direction of linear velocity of a body moving in a circle. Derive the relevant expression for it in terms of linear and angular velocities.

ANSWER: Acceleration due to the change in direction of linear velocity of a body moving in a circle is known as centripetal acceleration.

ANSWER:

DERIVATION

Consider an object moving in a circular path at constant speed. The direction of the instantaneous velocity is shown in figure as v_1 and v_2 at two points along the path and the distance between center and circular path is r . Acceleration is in the direction of the change in velocity, which

points directly toward the centre of rotation the center of the circular path. It is observed that the triangle formed by the velocity vectors and the one formed by the radii r and Δs are similar.

Both the triangles ABC and PQR are isosceles triangles. The two equal sides of the velocity vector triangles are the speeds $v_1 = v_2 = v$. Using the properties of two identical triangles we get:

$$\frac{\Delta v}{v} = \frac{\Delta s}{r}$$

$$\Delta v = \frac{v}{r} \Delta s$$

On dividing both sides by Δt we get

$$\frac{\Delta v}{\Delta t} = \frac{v}{r} \frac{\Delta s}{\Delta t}$$

$$\therefore a_c = \frac{\Delta v}{\Delta t}, \quad \text{and,} \quad v = \frac{\Delta s}{\Delta t}$$

$$a_c = \frac{v}{r} (v)$$

$$a_c = \frac{v^2}{r}$$

For centripetal acceleration in terms of angular velocity substituting the value of linear velocity ($v = r\omega$) in the above equation, we get

$$a_c = \frac{(r\omega)^2}{r}$$

$$a_c = \frac{r^2 \omega^2}{r}$$

$$a_c = r\omega^2$$

5. a) Define simple harmonic motion. Using the expression of instantaneous velocity $v = \sqrt{\frac{K}{m} \sqrt{x_0^2 - x^2}}$ of a body executing S.H.M, prove that the total energy of the body is constant.

ANSWER:

SIMPLE HARMONIC MOTION

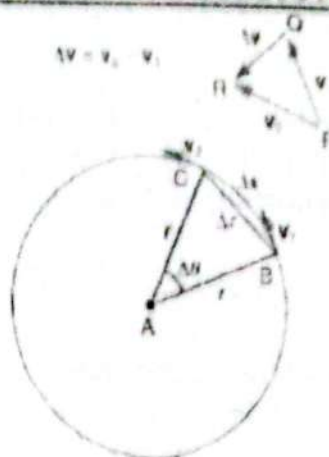
Simple harmonic motion is periodic motion that occurs whenever the restoring force is proportional to the displacement from equilibrium.

OR

Simple Harmonic Motion (SHM) is a type of oscillation or vibratory motion produced under the action of restoring force.

KINETIC ENERGY (K) OF THE OSCILLATOR:

Since the kinetic energy at any instant of the system is given by



Since the instantaneous velocity of the particle executing SHM is

$$v = \sqrt{\frac{K}{m} \sqrt{x_0^2 - x^2}}$$

On putting the above value in equation (i), we get

$$K = \frac{1}{2} m \left(\sqrt{\frac{K}{m} \sqrt{x_0^2 - x^2}} \right)^2$$

$$K = \frac{1}{2} m \left(\frac{K}{m} (x_0^2 - x^2) \right)$$

$$K = \frac{1}{2} k (x_0^2 - x^2)$$

The above equation represents the instantaneous kinetic energy of object executing simple harmonic motion.

Since the speed is maximum at equilibrium position i.e. at $x = 0$. Therefore above equation for maximum kinetic energy of the object at equilibrium point is

$$K = \frac{1}{2} k (x_0^2 - 0^2)$$

$$K = \frac{1}{2} k x_0^2$$

As the object is instantaneously at rest on extreme position, where $x = x_0$.

Therefore;

$$K = \frac{1}{2} k (x_0^2 - x_0^2)$$

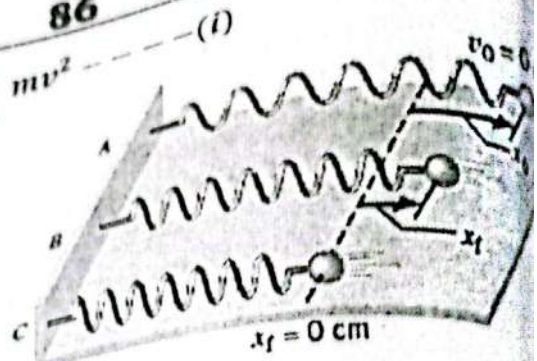
$$K = 0$$

POTENTIAL ENERGY (U) OF THE OSCILLATOR:

The figure suggests that the net force on the oscillator at O is $F_o = 0$ and at extreme point i.e. at A , it is $F_A = kx$. The average applied force exerted on the system in displacing it from O to A is

$$F_{av} = \frac{F_o + F_A}{2}$$

$$F_{av} = \frac{0 + kx}{2}$$



$$F_{av} = \frac{1}{2} kx$$

The work done in moving the object from O to A , against the elastic restoring force

$$W = F_{av} \cdot x = \frac{1}{2} kx^2$$

This work is stored in the spring-mass system as its elastic potential energy, therefore

$$U = \frac{1}{2} kx^2$$

The above equation expresses the instantaneous elastic potential energy of the object executing simple harmonic motion.

Mathematically, in general and practically above equation depend upon x , the instantaneous displacement. Hence the potential energy shall be maximum at $(x \pm x_0)$ are the extreme positions.

$$U_{max} = \frac{1}{2} kx_0^2$$

and minimum at O where $x = 0$

TOTAL ENERGY (E) OF THE OSCILLATOR:

Since the total mechanical energy of the system at any instant shall remain constant.

$$E = \text{Kinetic energy (K)} + \text{Potential energy (U)}$$

$$\therefore K = \frac{1}{2} k(x_0^2 - x^2), \quad U = \frac{1}{2} kx^2$$

$$E = \frac{1}{2} k(x_0^2 - x^2) + \frac{1}{2} kx^2$$

$$E = \frac{1}{2} k(x_0^2 - x^2 + x^2)$$

$$E = \frac{1}{2} kx_0^2$$

The above relations show that the elastic potential energy is zero where the displacement is zero and maximum at extreme position. Contrary to P.E, kinetic energy is maximum at zero displacement i.e. at equilibrium position and minimum at extreme positions but the total energy of a body executing SHM at any point is constant.

Q) What is Doppler Effect? Derive mathematical expression for apparent frequency heard by listener when:

(i) Listener moves towards stationary source

(ii) Source moves towards stationary listener

ANSWER:

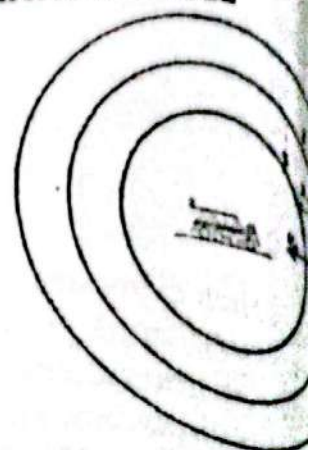
DOPPLER EFFECT

The Doppler Effect is a phenomenon in physics that describes the perceived change in frequency of a wave (such as sound or light) when the source of

the wave or the observer is in relative motion. In the case of sound the Doppler Effect occurs when there is relative motion between the of the sound and the observer.

i) LISTENER MOVES TOWARDS STATIONARY SOURCE

A stationary source emits a sound wave at frequency f_s and wavelength $\lambda_s = \frac{v}{f_s}$, where v is the speed of sound. An observer moving towards the stationary source with speed v_o would observe a shorter time interval between crests. Just as crest 1 reaches the observer, the next crest 2 is at distance λ_s ahead. Crest 2 catches up with the observer with time T' earlier than the time T_s , when the distance vT' the wave crest 2 travels toward the observer is equal to the wavelength of, minus the distance the observer travels towards the wave crest 2 in figure



$$\begin{aligned} vT' &= \lambda_s - v_o T' \\ vT' + v_o T' &= \lambda_s \\ \lambda_s &= (v + v_o)T' \end{aligned}$$

Given that

$$\begin{aligned} T' &= \frac{1}{f'''} \quad \text{and} \quad \lambda_s = \frac{v}{f_s} \\ \frac{v}{f_s} &= (v + v_o) \frac{1}{f'''} \\ f''' &= \frac{(v + v_o)}{v} f_s \end{aligned}$$

ii) SOURCE MOVES TOWARDS STATIONARY LISTENER

If the source is moving at velocity v_s toward a stationary observer on the right, the first figure shows that the wavelength, the distance between the crests is smaller to the right. The second Figure shows that at the instant that crest 4 is emitted, crest 3 has travelled



outward a distance vT_s from point 3. During the same time interval, the source has advanced a distance $v_s T_s$. The wavelength λ as measured by the observer on the right is the distance between crests 4 and 3:

$$\lambda = vT_s - v_s T_s$$

The frequency at which the crests arrive at the observer is the observed wave frequency f' . The observed period T_o

between the arrival of two crests is the time it takes sound to travel a distance $(v - v_s) T_s$:

$$T_o = \frac{(v - v_s) T_s}{v}$$

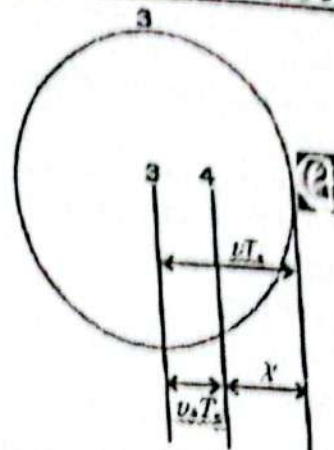
The observed frequency is

$$f' = \frac{1}{T_o} = \frac{1}{\frac{(v - v_s) T_s}{v}}$$

$$f' = \frac{v}{(v - v_s) T_s}$$

$$f' = \frac{v}{(v - v_s)} \frac{1}{T_s}$$

$$f' = \frac{v}{(v - v_s)} f_s$$



PHYSICS 2024

(Science General Group)

TIME: 3 Hours

(85 Marks)

SECTION 'A' (Multiple Choice Questions)(17)

NOTE: i) This section consists of 17 part questions and all are to be answered. Each question carries 1 mark.

ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

iii) The use of scientific calculator is allowed. All notations are used in their usual meanings

1. Select the correct answer for each from the given options:

i) A sky diver falls through the air with terminal velocity. The force of air resistance on him is:

* half of his weight

* equal to his weight ✓

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* zero

* twice of his weight

* infinite ✓

ii) The resistance of superconductor is:

* Zero ✓

* Finite

* Change with every conductor

* extreme position

iii) In SHM kinetic energy is maximum at:

* mean & extreme positions

* mean position ✓

* $1/2 CV$

* all positions

* CV^2

iv) Energy stored in the capacitor is equal to:

* friction ✓

* $\frac{1}{2} CV^2$ ✓

* thrust

v) This force is called as self-adjusting force:

* tension

* weight

* $L^2 T^{-1}$

vi) The dimensions of angular momentum are given by:

* $ML^{-1} T^{-2}$

* $ML^{-1} T^{-1}$

* Optical fiber

vii) A communication channel does not consist of:

* Receiving equipment

* Transmission line

* 10N

* Free space ✓

* 7N

viii) A body suspended by a weighing scale weighs 10N out of the water. This is the buoyant force on the body when submerged in water.

* 5N

ix) This pair of angles a projectile will obtain the equal horizontal range for a given velocity:

* $(23^\circ, 58^\circ)$

* $(18^\circ, 74^\circ)$

* $(37^\circ, 70^\circ)$

* $(33^\circ, 57^\circ)$ ✓

* $(23^\circ, 58^\circ)$

x) A jet engine plane develops a forward force of 2000N. The velocity of flying at 200 m/s. The power of the engine will be:

* $4 \times 10^5 \text{ hp}$

* $4 \times 10^5 \text{ W}$

* $4 \times 10^5 \text{ kW}$ ✓

xi) A mass spring oscillator has a time period T, if the mass is the time period will become:

* $4 \times 10^5 \text{ MW}$

* T

* $2T$

* $\sqrt{2}T$ ✓

* $T/\sqrt{2}$

xii) A charge of 5C is moved from one point to another point in an electric field and 1000J of work is done by the field. The potential difference between the two points will be:

* 100V

* 200V ✓

* 500V

* 800V

xiii) Electric resistance of a conductor does not depend upon:

* Length of conductor

* Temperature of conductor

* Area of cross-section of conductor

* Potential difference between the sides of the conductor

Q) This electromagnetic wave in the given option has the shortest wavelength:

Radio waves

* ultraviolet waves ✓

microwaves

* infrared waves

Q) The locus of all points in the same phase of vibration is called:

Wave number

* wave pulse

wavelength

* wave front ✓

Q) The maximum number of beats that can be heard by human ear:

* 5

* 7 ✓

* 9

Q) A body moving along the circular path may have constant:

Velocity

* Acceleration

* Momentum

* Speed ✓

SECTION 'B' (Short-Answer Questions) (36)

NOTE: Answer any NINE part questions from this section. All questions carry equal marks. Draw diagrams where necessary. Use of scientific calculator is allowed. All notations are used in their usual meanings.

Q) Prove that following equations are dimensionally correct:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$S = v_i t + \frac{1}{2} a t^2$$

ANSWER:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

PROOF:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

substituting the dimensions of various quantities in the above equation we get

$$T = 2\pi \left(\frac{L}{LT^{-2}} \right)^{\frac{1}{2}}$$

In the above equation 2π is dimensionless constant hence therefore

$$T = (L^{1-1} T^{-2})^{\frac{1}{2}} = (L^0 T^{-2})^{\frac{1}{2}}$$

$$T = T$$

Each term on both sides of the above equation has same dimensions. Hence the given expression is dimensionally correct.

$$S = v_i t + \frac{1}{2} a t^2$$

PROOF:

$$S = v_i t + \frac{1}{2} a t^2$$

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On substituting the dimensions of various quantities in the above we get

$$L = (LT^{-1})(T) + \frac{1}{2}(LT^{-2})(T)^2$$

In the above equation $\frac{1}{2}$ is dimensionless constant hence therefore

$$L = LT^{-1+1} + LT^{-2+2} = LT^0 + LT^0$$

$$L = L + L$$

Each term on both sides of the above equation has same dimension. Proved the given expression is dimensionally correct.

ii) A 50g bullet is fired into a 10kg block that is suspended by a string so that it can swing as a pendulum. If the block is displaced so that the height of gravity rises by 10cm, find the initial velocity of bullet.

ANSWER:

DATA:

$$\text{Mass of bullet} = m = 50g = \frac{50}{1000} = 0.05kg$$

$$\text{Mass of block} = M = 10kg$$

$$\text{Height} = h = 10cm = \frac{10}{100} = 0.1m$$

$$\text{initial speed of the bullet} = u_1 = ?$$

SOLUTION:

Applying law of conservation of momentum

$$Mu_1 + mu_2 = Mv_1 + mv_2$$

$$(10)(0) + (0.05)u_2 = (10)V + (0.05)V$$

$$0.05u_2 = 10.05V$$

$$u_2 = \frac{10.05V}{0.05} = 201V \text{ --- (1)}$$

Since kinetic energy of bullet and block system is:

$$K.E. = \frac{1}{2}MV^2 = mgh$$

$$\frac{1}{2}V^2 = (9.8)(0.1)$$

$$V^2 = 2(0.98) = 1.96$$

$$V = 1.4m/s$$

Substituting the value of V in equation 1

$$u_2 = 201(1.4)$$

$$u_2 = 281.4m/s$$

iii) Define escape velocity. Derive the expression for escape velocity from earth's surface.

ANSWER: (See Question No. 2(vi), 2024 P.E. & P.M.)

iv) Calculate the viscous drag on a drop of oil of 0.1mm diameter falling through air at its terminal velocity.

viscosity of air $1.8 \times 10^{-5} \text{ Pa-s}$, density of oil $= 850 \text{ kg/m}^3$).

ANSWER: (See Question No. 2(x), 2024 P.E. , P.M.)

Define Electric flux, give its SI unit. Under what condition the electric flux through a surface will be:

a) maximum b) minimum

ANSWER:

ELECTRIC FLUX

The word flux is derived from the Latin "Fluxus", which means flow.

The dot product of the electric field $\vec{E}$ and vector area $\vec{A}$ is called the electric flux and denoted by Φ_e .

OR

The electric flux (Φ_e) of an electric field ($\vec{E}$) is a measure of the number of electric field lines passing through an area ($\vec{A}$).

The greater the area, the more field lines pass through it, and thus the greater the flux. Mathematically

$$\Phi_e = \vec{E} \cdot \vec{A}$$

where vector area A is parallel to electric field lines.

UNIT

The SI unit of electric flux is $\text{N.m}^2/\text{C}$.

MAXIMUM ELECTRIC FLUX

The electric flux of a surface is maximum when the vector area $\vec{A}$ is parallel to the direction of the electric field, i.e.

$$\Phi_e = \vec{E} \cdot \vec{A} \cos 0^\circ$$

$$\Phi_e = \vec{E} \cdot \vec{A} (1)$$

$$\Phi_e = \vec{E} \cdot \vec{A}$$

MINIMUM ELECTRIC FLUX

The electric flux of a surface is minimum when the vector area $\vec{A}$ is perpendicular to the direction of the electric field, i.e.

$$\Phi_e = \vec{E} \cdot \vec{A} \cos 90^\circ$$

$$\Phi_e = \vec{E} \cdot \vec{A} (0)$$

$$\Phi_e = 0$$

Derive an expression for the equivalent capacitance of three unequal capacitors connected in series OR parallel.

ANSWER:

1) WHEN TWO CAPACITORS CONNECTED IN SERIES

Three capacitors are connected in series combination. When a potential V is applied across them, the same amount Q of charge will appear on each of the capacitor. The charges on capacitors connected in series are the same.

$$Q = Q_1 = Q_2 = Q_3$$

In general, the total potential difference across any number of capacitors connected in series is the sum of the potential differences across individual capacitors. i.e.

$$V = V_1 + V_2 + V_3$$

To find an expression for equivalent capacitance C_{eq} , find the potential difference across each actual capacitor and equivalent capacitor.

$$\therefore V = \frac{Q}{C}, \quad V_1 = \frac{Q}{C_1}, \quad V_2 = \frac{Q}{C_2} \text{ and, } V_3 = \frac{Q}{C_3}$$

$$\begin{aligned} \frac{Q}{C} &= \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} \\ \frac{Q}{C} &= Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) \\ \frac{1}{C} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \end{aligned}$$

This expression shows that the reciprocal of the equivalent capacitance is the algebraic sum of the reciprocal of the individual capacitances. The equivalent capacitance of a series combination is always less than the smallest individual capacitance.

(b) WHEN THREE CAPACITORS CONNECTED IN PARALLEL

Consider three capacitors are connected in parallel combination. The potential difference across each capacitor connected in parallel is same and equal to the potential difference applied across the combination. i.e.,

$$V = V_1 = V_2 = V_3$$

In general, when a potential difference V is applied across several capacitors connected in parallel, that potential difference V is same across each capacitor. The total charge Q stored on the equivalent capacitor is the sum of the charges stored on all the capacitors.

To find an expression for equivalent capacitance C_{eq} , find the total charge on the actual capacitor and equivalent capacitor.

$$\therefore Q = CV, \quad Q_1 = C_1V, \quad Q_2 = C_2V, \text{ and, } Q_3 = C_3V$$

The total charge on the parallel combination is given as

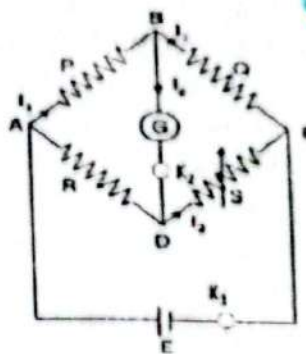
$$\begin{aligned} Q &= Q_1 + Q_2 + Q_3 \\ C_{eq}V &= C_1V + C_2V + C_3V \\ C_{eq} &= C_1 + C_2 + C_3 \end{aligned}$$

Therefore, the equivalent capacitance of a parallel combination of capacitors is the algebraic sum of the individual capacitances. It is always greater value than any of the individual capacitances.

vii) Derive an equation for balanced Wheatstone bridge.

ANSWER:

Wheatstone Bridge Principle states that if four resistance P , Q , R and S are arranged to form a bridge as shown in figure with a cell E and one key K_1 between the point A and C and galvanometer G and tapping key K_2 between the points B and D , closing K_1 first and K_2 later on, if the galvanometer shows no deflection then bridge is balanced.

**DERIVATION**

The Current distribution in the circuit is shown in the figure.

The total current by cell is I , it distributed to P and R as I_1 and $I - I_1$. The current through galvanometer is I_g

Current in Q is $I_1 - I_g$ and through S current is $I - I_1 + I_g$ and the resistance of galvanometer is G .

If we apply Kirchhoff's 2nd law in $ABDA$, we get

$$I_1 P + I_g G - (I - I_1) R = 0 \quad \text{--- (i)}$$

Now apply Kirchhoff's 2nd law in $BCDB$, we get

$$(I_1 - I_g) Q - (I - I_1 + I_g) S - I_g G = 0 \quad \text{--- (ii)}$$

If value of R is such that the galvanometer shows no deflection that is $I_g = 0$. Putting this value in equation (i) and (ii) we get

$$I_1 P + (0) G - (I - I_1) R = 0$$

$$I_1 P - (I - I_1) R = 0$$

$$I_1 P + I_1 R = IR$$

$$I_1 (P + R) = IR$$

$$\frac{P + R}{R} = \frac{I}{I_1} \quad \text{--- (iii)}$$

Also

$$(I_1 - 0) Q - (I - I_1 + 0) S - (0) G = 0$$

$$I_1 Q - (I - I_1) S = 0$$

$$I_1 Q + I_1 S = IS$$

$$I_1 (Q + S) = IS$$

$$\frac{Q + S}{S} = \frac{I}{I_1}$$

On comparing equation (iii) and (iv), we get

$$\frac{P + R}{R} = \frac{Q + S}{S}$$

$$PS + RS = QR + RS$$

$$PS = QR$$

$$\frac{P}{Q} = \frac{R}{S} \quad \text{Derived}$$

viii) Three resistors of 1Ω , 2Ω and 3Ω are connected in series. What is the total resistance of the combination? If the combination is connected to a battery of emf 24 V and negligible internal resistance. Find the potential drop across each resistor.

ANSWER:

DATA:

resistance of 1st resistor = $R_1 = 1\Omega$

resistance of 2nd resistor = $R_2 = 2\Omega$

resistance of 3rd resistor = $R_3 = 3\Omega$

Equivalent Resistance = $R = ?$

emf of battery = $\epsilon = 24\text{ V}$

Potential drop across 1st resistor = $V_1 = ?$

Potential drop across 2nd resistor = $V_2 = ?$

Potential drop across 3rd resistor = $V_3 = ?$

SOLUTION:

FOR EQUIVALENT RESISTANCE

$$\therefore R = R_1 + R_2 + R_3$$

$$R = 1 + 2 + 3$$

$$R = 6\Omega$$

FOR POTENTIAL DROP

$$\therefore \epsilon = IR$$

$$\frac{\epsilon}{R} = I$$

$$I = \frac{24}{6}$$

$$I = 4\text{ A}$$

Now

$$V_1 = I_1 R_1$$

$$V_1 = (4)(1)$$

$$V_1 = 4\text{ V}$$

$$V_2 = I_2 R_2$$

$$V_2 = (4)(2)$$

$$V_2 = 8\text{ V}$$

$$V_3 = I_3 R_3$$

$$V_3 = (4)(3)$$

$$V_3 = 12\text{ V}$$

ix) If the speed of sound in air at 27°C is 345 m/s . Find the speed of sound at 127°C .

ANSWER:

DATA:

First Temperature = $T_o = 27^\circ\text{C} = 27 + 273 = 300\text{ K}$

Second Temperature = $T = 127^\circ\text{C} = 127 + 273 = 400\text{ K}$

Speed of sound at $27^\circ\text{C} = v_o = 345\text{ m/s}$

Speed of sound at $127^\circ\text{C} = v_t = ?$

SOLUTION:

$$\frac{v_t}{v_o} = \sqrt{\frac{T}{T_o}}$$

$$v_t = \sqrt{\frac{T}{T_0}} \times v_0$$

$$v_t = \sqrt{\frac{400K}{300K}} \times 345$$

$$v_t = 1.1547 \times 345$$

$$v_t = 398.37 \text{ ms}^{-1}$$

How many lines per centimeter are there in a grating which gives 1st order spectrum at an angle of 30° when the wavelength of light is $6 \times 10^{-5} \text{ cm}$

SOLUTION:

DATA:

Order of spectra = $m = 1$

Angle of spectra = $\theta = 30^\circ$

Wavelengths of light = $\lambda = 6 \times 10^{-5} \text{ cm}$

Number of lines = $N = ?$

SOLUTION:

$$\therefore d \sin \theta = \lambda m$$

$$d = \frac{\lambda m}{\sin \theta}$$

$$d = \frac{(6 \times 10^{-5})(1)}{\sin 30^\circ}$$

$$d = \frac{6 \times 10^{-5}}{0.5}$$

$$d = 1.2 \times 10^{-4} \text{ cm}$$

$$\therefore N = \frac{1}{d}$$

$$N = \frac{1}{1.2 \times 10^{-4}}$$

$$N = 8333.33 \text{ lines per cm}$$

A helicopter is ascending at a rate of 12 m/s . At a height of 80 m above the ground, a packet is dropped. Find the time taken by packet to reach the ground.

SOLUTION:

DATA:

Ascending rate = $V_i = 12 \text{ ms}^{-1}$

Height = $-h = 80 \text{ m}$

Acceleration due to gravity = $g = -9.8 \text{ ms}^{-2}$

SOLUTION:

$$\therefore S = V_i t + \frac{1}{2} a t^2$$

Case

$$\begin{aligned} \therefore h &= v_i t + \frac{1}{2} g t^2 \\ -80 &= (12)t + \frac{1}{2} (-9.8) t^2 \\ -80 &= 12t - 4.9t^2 \\ 4.9t^2 - 12t - 80 &= 0 \end{aligned}$$

By using Quadratic formula

$$\begin{aligned} \therefore x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ a &= 4.9, \quad b = -12, \quad c = -80 \\ t &= \frac{-(-12) \pm \sqrt{(-12)^2 - 4(4.9)(-80)}}{2(4.9)} \\ t &= \frac{12 \pm \sqrt{144 + 1568}}{9.8} = \frac{12 \pm \sqrt{1712}}{9.8} \\ t &= \frac{12 + 41.37}{9.8} \end{aligned}$$

$$\begin{aligned} t &= \frac{12 + 41.37}{9.8} \\ t &= \frac{53.376}{9.8} \\ t &= 5.44 \text{ sec} \end{aligned}$$

$$\begin{aligned} t &= \frac{12 - 41.37}{9.8} \\ t &= \frac{-29.376}{9.8} \\ t &= -2.99 \end{aligned}$$

Neglect this value because
and time never be negative

The package will take 5.4 seconds to reach the ground.
xii) State Pascal's law. Give the working of hydraulic lift.

ANSWER:

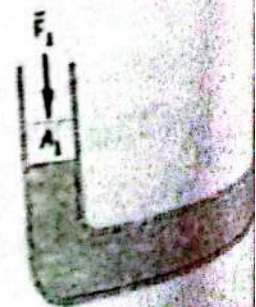
PASCAL'S PRINCIPLE

Pascal's principle states that "when a change in pressure is applied to any static fluid, it is transmitted perfectly to all parts of the fluid and to the wall of the container".

HYDRAULIC LIFT:

Hydraulic lift is able to raise up large weight up to relatively distance.

The hydraulic lift, or jack is applications of hydraulics being used as a simple machine to multiply force. It contains an incompressible fluid in a U-shaped tube which is narrower at start and becomes wider area at end



which is fixed with a movable piston on each side. If a small force F_1 is applied to the small piston of the hydraulic lift as shown in figure, the pressure is transmitted with in all directions. The pressure on the large piston is the same as the pressure on the small piston; however, the force F_2 on the large piston is greater because of its large surface area which used to uplift the car.

xiii) A source of sound and listener are moving towards each other with velocities 0.5 and 0.2 times the speed of sound respectively. If the source is emitting 2kHz tone, calculate the frequency heard by the listener. (Speed of sound = 340 m/s)

ANSWER:

DATA:

Velocity of source = $v_s = 0.5v$

Velocity of listener = $v_o = 0.2v$

Frequency emitted by source = $f_s = 2\text{kHz} = 2000\text{Hz}$

Frequency heard by the listener = $f_o = ?$

SOLUTION:

$$\therefore f_o = \left(\frac{v + v_o}{v - v_s} \right) f_s$$

$$f_o = \left(\frac{v + 0.2v}{v - 0.5v} \right) \times 2000\text{Hz}$$

$$f_o = \left(\frac{1.2v}{0.5v} \right) \times 2000\text{Hz}$$

$$f_o = 2.4 \times 2000\text{Hz}$$

$$f_o = 4800\text{Hz}$$

v) Derive an expression for centripetal acceleration $a_c = \frac{v^2}{r}$

ANSWER: (See Question No. 4(b), 2024 P.E. & P.M.)

SECTION "C" (Detailed Answer Questions)(32)

NOTE: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

a) Two vectors $\vec{A}_1$ and $\vec{A}_2$ making angles θ_1 and θ_2 with the horizontal. Describe addition of these vectors by rectangular component method also give formula for magnitude of resultant vector and its direction.

ANSWER:

ADDITION OF VECTORS BY RECTANGULAR COMPONENTS

METHOD:

Consider two vectors $\vec{A}_1$ and $\vec{A}_2$ making angles θ_1 and θ_2 with +ve x-axis respectively.

Resolve vector $\vec{A}_1$ into two rectangular components $\vec{A}_{1x}$ and $\vec{A}_{1y}$ as shown in figure

Magnitude of these components are:

$$A_{1x} = A_1 \cos \theta_1$$

And

$$A_{1y} = A_1 \sin \theta_1$$

Resolve vector $\vec{A}_2$ into two rectangular component A_{2x} and A_{2y} as shown in figure

Magnitude of these components are:

$$A_{2x} = A_2 \cos \theta_2$$

And

$$A_{2y} = A_2 \sin \theta_2$$

Now move vector $\vec{A}_2$ parallel to itself so that its initial point (tail) lie the terminal point (head) of vector $\vec{A}_1$ as shown in the figure

Representative lines of $\vec{A}_1$ and $\vec{A}_2$ are OA and OB respectively. Join B which is equal to resultant vector $\vec{A}_1$ and $\vec{A}_2$

Resultant vector along x-axis can be determined as:

$$\overline{OD} = \overline{OC} + \overline{CD}$$

$$\overline{OD} = \overline{OC} + \overline{AE}$$

$$\vec{R}_x = \vec{A}_{1x} + \vec{A}_{2x}$$

$$\vec{R}_x = A_1 \cos \theta_1 + A_2 \cos \theta_2$$

Resultant vector along y-axis can be determined as:

$$\overline{DB} = \overline{CA} + \overline{EB}$$

$$\overline{OD} = \overline{DE} + \overline{EB}$$

$$\vec{R}_y = \vec{A}_{1y} + \vec{A}_{2y}$$

$$\vec{R}_y = A_1 \sin \theta_1 + A_2 \sin \theta_2$$

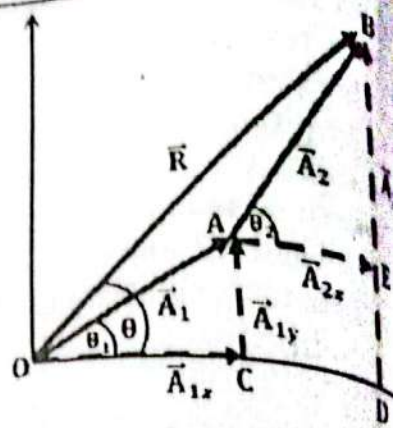
Now we will determine the magnitude of resultant vector by using Pythagoras' theorem.

$$\text{Hypotenous}^2 = \text{Base}^2 + \text{Perpendicular}^2$$

$$|\overline{OB}|^2 = |\overline{OD}|^2 + |\overline{DB}|^2$$

$$R^2 = R_x^2 + R_y^2$$

$$R = \sqrt{R_x^2 + R_y^2}$$



$$R = \sqrt{(A_1 \cos \theta_1 + A_2 \cos \theta_2)^2 + (A_1 \sin \theta_1 + A_2 \sin \theta_2)^2}$$

Finally the direction of resultant vector will be determined.

Again in the right angled triangle $\triangle DOB$:

$$\tan \angle DOB = \frac{DB}{OD}$$

$$\tan \angle DOB = \tan \theta = \frac{R_y}{R_x}$$

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$$

Where θ is the angle that the resultant vector makes with the positive x -axis. In this way we can add a number of vectors in a very easy manner. This method is known as addition of vectors by rectangular components method.

b) Define simple Harmonic motion, "A particle is moving in a circle with constant speed, prove that its projection execute simple harmonic motion on one of the diameter of the circle."

ANSWER:

SIMPLE HARMONIC MOTION

Simple harmonic motion is periodic motion that occurs whenever the restoring force is proportional to the displacement from equilibrium.

OR

Simple Harmonic Motion (SHM) is a type of oscillation or vibratory motion produced under the action of restoring force.

PROOF:

Consider a particle 'P' moving around a circle of radius 'r' ($r = x_0$) with uniform angular velocity ' ω '. The linear speed of the particle is

$$V = x_0 \omega$$

Since 'P' is moving along the circular path, its acceleration is equal to centripetal acceleration which is directed towards the centre of circle.

$$a_p = a_c$$

But

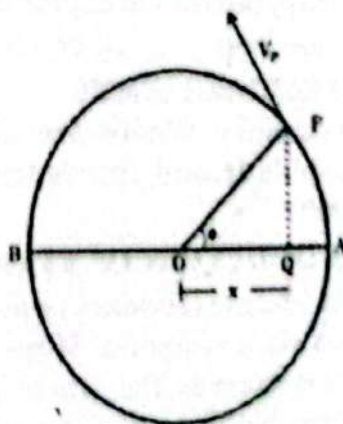
$$a_c = \frac{v^2}{r}$$

And $r = x_0$

$$a_p = \frac{v^2}{x_0} \times \frac{x_0}{x_0}$$

$$a_p = x_0 \times \frac{v^2}{x_0^2}$$

$$a_p = x_0 \omega^2$$



The acceleration of the projection 'a_Q' is equal to the component of acceleration of particle 'P' along the diameter AB
In a Δ POQ

$$\frac{OQ}{OP} = \cos \theta$$

$$\frac{x}{x_0} = \cos \theta$$

$$\frac{a_Q}{a_P} = \cos \theta$$

$$a_Q = -a_P \cos \theta$$

(Negative sign shows that, the acceleration of the body executing SHM is directed towards its mean position).

$$\text{But } a_P = x_0 \omega^2$$

$$a_Q = -x_0 \omega^2 \left(\frac{x}{x_0} \right)$$

$$a_Q = -x \omega^2$$

Since the particle "P" moving with a uniform angular velocity the "ω" is a constant.

$$a_Q \propto -x$$

Above result shows that, the acceleration of the projection is proportional to its displacement and always directed towards its mean position, therefore the motion of the projection of a particle moving in a circular path with constant speed is simple harmonic motion.

4. a) What is an electric dipole? Derive the expression for electric field intensity due to an electric dipole at a point which is at a perpendicular distance 'y' from the centre of the dipole.

ANSWER: (See Question No. 3(b), 2024 P.E. & P.M.)

b) What is stationary waves? Stationary waves are established in a string, derive the expressions for the frequencies if the string is vibrating in

- i) one loop ii) two loops iii) three loops iv) n loops

STANDING WAVE

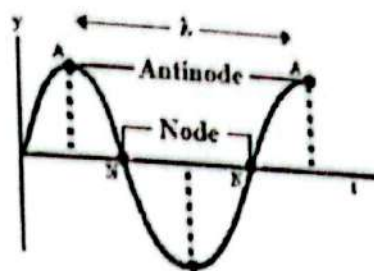
Stationary Waves are the combination of two waves having same amplitude and frequency but moving in opposite directions in a string.

PRODUCTION OF STANDING WAVE

Standing (Stationary) waves occur when a string is tightly stretched between two rigid supports. If the string is plucked from the half of its length, a crest extends the whole distance between the supports. This is clearly half the wavelength of the transverse wave produced in the string. This wave is reflected at the boundary and the reflected wave interferes with the incident wave so that the wave appears to stand.

NODES AND ANTINODES

points at which the string vibrates with minimum or zero amplitude are called nodes (N) whereas the points at which string vibrates with maximum amplitude are called antinode (A).



EXPRESSION FOR THE FREQUENCY:

Consider a string of length 'L' be stretched and fixed by two ends. Let 'T' be the tension in the string. If the string is plucked at its center point, two transverse waves start from the centre and move in the opposite direction. Due to this motion of waves string vibrates in one loop.

WHEN STRING VIBRATING IN ONE LOOPS:

If the string is plucked from the one-half of its length, string vibrates in one loop. Frequency of Wave when a string vibrates in one loop is given by:

$$V = f_1 \lambda_1$$

$$\because L = \frac{\lambda_1}{2} \Rightarrow \lambda_1 = 2L, \quad V = \sqrt{\frac{TL}{M}}$$

$$\sqrt{\frac{TL}{M}} = f_1(2L)$$

$$\sqrt{\frac{TL}{M}} = 2Lf_1$$

$$f_1 = \frac{1}{2L} \sqrt{\frac{TL}{M}}$$

$$f_1 = \frac{1}{2L} \sqrt{\frac{T}{\frac{M}{L}}}$$

μ is the mass per unit length (linear density), $\mu = \frac{M}{L}$ then the above equation becomes

$$f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

WHEN STRING VIBRATING IN TWO LOOPS:

If the string is plucked from the one-fourth of its length, string vibrates in two loops. Frequency of Wave when a string vibrates in two loops is given by:

$$V = f_2 \lambda_2$$

$$\therefore L = \frac{\lambda_2}{2} + \frac{\lambda_2}{2} = \lambda_2, \quad V = \sqrt{\frac{TL}{M}}$$

$$\sqrt{\frac{TL}{M}} = f_2(L)$$

$$\sqrt{\frac{TL}{M}} = L f_2$$

$$f_2 = \frac{1}{L} \sqrt{\frac{TL}{M}}$$

$$f_2 = \frac{1}{L} \sqrt{\frac{T}{\mu}}$$

$$f_2 = 2 \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

$$f_2 = 2f_1$$

$$f_1 = \frac{V}{2L} \Rightarrow 2f_1 = \frac{V}{L}$$

Or

$$f_2 = 2f_1$$

When a String is Vibrates in two Loops, the frequency of vibration is second harmonic or first overtone.

WHEN STRING VIBRATING IN THREE LOOPS:

If the string plucked from the one-sixth of its length, string vibrates in three loops. Frequency of Wave when a string vibrates in three loops

$$V = f_3 \lambda_3$$

$$\therefore L = \frac{\lambda_1}{2} \Rightarrow \lambda_1 = 2L, \quad V = \sqrt{\frac{TL}{M}}$$

$$\therefore L = \frac{\lambda_3}{2} + \frac{\lambda_3}{2} + \frac{\lambda_3}{2} = \frac{3\lambda_3}{2} \Rightarrow \lambda_3 = \frac{2L}{3}, \quad V =$$

$$V = f_3 \left(\frac{2L}{3} \right)$$

$$v = f_3 \frac{2L}{3}$$

$$f_3 = \frac{3v}{2L}$$

Or

$$f_3 = 3 \left(\frac{v}{2L} \right)$$

$$f_3 = \frac{3}{2L} \sqrt{\frac{TL}{M}}$$

$$f_3 = 3 \left(\frac{1}{2L} \sqrt{\frac{TL}{\mu}} \right)$$

$$f_3 = 3f_1$$

When a String is Vibrates in three Loops, the frequency of vibration is called third harmonic or second overtone.

5. a) Two smooth, rigid and not-rotating spheres of masses m_1 and m_2 , moving with initial velocities U_1 and U_2 , collide elastically in one dimension. Derive the expressions for the final velocities of the spheres after the collision.

DERIVATION:

Consider two smooth, rigid and non-rotating spheres of masses m_1 and m_2 , initially moving with the velocities U_1 and U_2 respectively in one dimension, collide elastically. After collision their velocities will be V_1 and V_2 .

Applying Law of conservation of momentum we get:

Total momentum before collision

= Total momentum after collision

$$m_1 U_1 + m_2 U_2 = m_1 V_1 + m_2 V_2$$

$$m_1 U_1 - m_1 V_1 = m_2 V_2 - m_2 U_2$$

$$m_1 (U_1 - V_1) = m_2 (V_2 - U_2) \text{ --- (i)}$$

Applying Law of conservation of kinetic energy we get:

$$\frac{1}{2} m_1 U_1^2 + \frac{1}{2} m_2 U_2^2 = \frac{1}{2} m_1 V_1^2 + \frac{1}{2} m_2 V_2^2$$

$$\frac{1}{2} m_1 U_1^2 - \frac{1}{2} m_1 V_1^2 = \frac{1}{2} m_2 V_2^2 - \frac{1}{2} m_2 U_2^2$$

$$\frac{1}{2} m_1 (U_1^2 - V_1^2) = \frac{1}{2} m_2 (V_2^2 - U_2^2)$$

$$\frac{1}{2} m_1 (U_1^2 - V_1^2) = \frac{1}{2} m_2 (V_2^2 - U_2^2)$$

$$m_1 (U_1 - V_1)(U_1 + V_1) = m_2 (V_2 - U_2)(V_2 + U_2) \text{ --- (ii)}$$

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FINAL VELOCITY OF 1ST BODY 'V₁'

Equation (2) divided by equation (1).

$$\frac{m_1(U_1 - V_1)(U_1 + V_1)}{m_1(U_1 - V_1)} = \frac{m_2(V_2 - U_2)(V_2 + U_2)}{m_2(V_2 - U_2)}$$

$$(U_1 + V_1) = (V_2 + U_2) \quad \text{--- (iii)}$$

On substituting the value of V₂ in equation (i) we get:

$$m_1(U_1 - V_1) = m_2\{(U_1 + V_1 - U_2) - U_2\}$$

$$m_1(U_1 - V_1) = m_2(U_1 + V_1 - 2U_2)$$

$$m_1U_1 - m_1V_1 = m_2U_1 + m_2V_1 - 2m_2U_2$$

$$m_1V_1 + m_2V_1 = m_1U_1 - m_2U_1 + 2m_2U_2$$

$$V_1(m_1 + m_2) = U_1(m_1 - m_2) + 2m_2U_2$$

$$V_1 = \frac{U_1(m_1 - m_2) + 2m_2U_2}{(m_1 + m_2)}$$

$$V_1 = \frac{U_1(m_1 - m_2)}{(m_1 + m_2)} + \frac{2m_2U_2}{(m_1 + m_2)}$$

FINAL VELOCITY OF 2ND BODY 'V₂'

On substituting the value of V₁ in equation (iii) we get:

$$V_2 = U_1 + \left\{ \frac{U_1(m_1 - m_2)}{(m_1 + m_2)} + \frac{2m_2U_2}{(m_1 + m_2)} \right\} - U_2$$

$$V_2 = U_1 + \frac{U_1(m_1 - m_2)}{(m_1 + m_2)} + \frac{2m_2U_2}{(m_1 + m_2)} - U_2$$

$$V_2 = U_1 \left\{ 1 + \frac{(m_1 - m_2)}{(m_1 + m_2)} \right\} + U_2 \left\{ \frac{2m_2}{(m_1 + m_2)} - 1 \right\}$$

$$V_2 = U_1 \left\{ \frac{(m_1 + m_2) + (m_1 - m_2)}{(m_1 + m_2)} \right\} + U_2 \left\{ \frac{2m_2 - (m_1 + m_2)}{(m_1 + m_2)} \right\}$$

$$V_2 = U_1 \left\{ \frac{2m_1}{(m_1 + m_2)} \right\} + U_2 \left\{ \frac{m_2 - m_1}{(m_1 + m_2)} \right\}$$

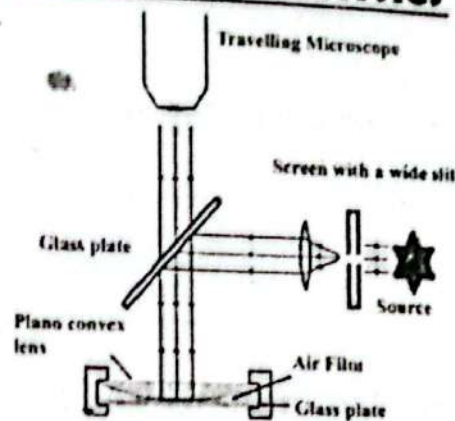
$$V_2 = \frac{2m_1U_1}{(m_1 + m_2)} + \frac{U_2(m_2 - m_1)}{(m_1 + m_2)} \quad \text{Derived}$$

b) What are Newton's rings? Explain the process of their formation the expressions for the radius of nth bright and dark rings.

NEWTON'S RINGS AND ITS FORMATION

If a Plano convex lens of radius of curvature R is placed on a flat glass plate. The air gap between the two surfaces will increase from zero at the point of contact of lens and flat glass plate to the edge of the lens.

figure). If the setup is illuminated by monochromatic light of wavelength λ incident normally on the Plano convex lens. Due to the curvature of lens the air film between the lens and glass plate is also spherical. The interference pattern we obtain is in the form of alternate dark and bright concentric rings called Newton's rings. The center of the rings is the point where air film thickness is zero.



Thomas Young figured out that the central ring of the interference pattern is a dark spot due to zero path difference and phase reversal of 180° between the rays reflected from the upper and lower surfaces.

DERIVATION

According to geometrical theorem, the product of intercepts of intersecting chord is equal to the product of sections of diameter.

$$DB \times BE = AB \times BC$$

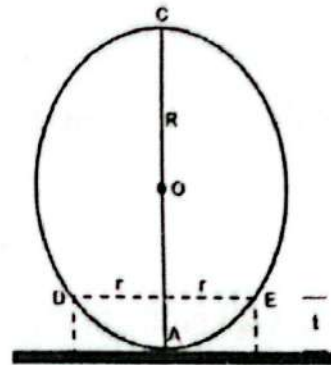
$$r \times r = t(2R - t)$$

$$r^2 = 2Rt - t^2$$

As "t" is very small then t^2 will be so small which may be neglected, then

$$r^2 = 2Rt$$

$$t = \frac{r^2}{2R}$$



FOR BRIGHT RING

The condition for constructive interference in thin film is,

$$2tn = \left(m + \frac{1}{2}\right)\lambda$$

Where $m = 0, 1, 2, \dots$

Putting the value of "t" in the above equation we get,

$$2\left(\frac{r^2}{2R}\right)n = \left(m + \frac{1}{2}\right)\lambda$$

Since $n = 1$ for air film

$$\left(\frac{r^2}{R}\right)(1) = \left(m + \frac{1}{2}\right)\lambda$$

$$r^2 = \left(m + \frac{1}{2}\right)\lambda \times R$$

Taking square root on both sides

$$r = \sqrt{\left(m + \frac{1}{2}\right) R \lambda}$$

For 1st bright ring, $m = 0$

$$r = \sqrt{\left(0 + \frac{1}{2}\right) \lambda \times R}$$

$$r = \sqrt{\frac{R \lambda}{2}}$$

For 2nd bright ring, $m = 1$

$$r = \sqrt{\frac{3 R \lambda}{2}}$$

For 3rd bright ring, $m = 2$

$$r = \sqrt{\frac{5 R \lambda}{2}}$$

Similarly, for Nth bright ring,

$$m = N - 1$$

$$r_N = \sqrt{\left(N - 1 + \frac{1}{2}\right) R \lambda}$$

Radius of nth Bright ring

$$r_N = \sqrt{\left(N - \frac{1}{2}\right) R \lambda}$$

FOR DARK RING

The condition for destructive interference in thin film is,

$$2t = m\lambda$$

Putting the value of "t" in the above equation we get,

$$2 \left(\frac{r^2}{2R} \right) = m\lambda$$

$$r^2 = Rm\lambda$$

Taking square root on both sides

$$r = \sqrt{Rm\lambda}$$

Where $m = 0, 1, 2, \dots$

$$r_m = \sqrt{Rm\lambda} \text{ Derived}$$

YOUSUIFS MODEL QUESTION PAPER 1

For Annual Examination 2024 and Onwards

PHYSICS (THEORY)**CLASS XI (SCIENCE)**

Total Time: 3 Hours

Total Marks: 85

SECTION 'A' (COMPULSORY)**MULTIPLE CHOICE QUESTIONS (M.C.Qs)**

NOTE: This Section consists of 17 MCQs and all are to be answered.
Each MCQ carry 1 marks.

Choose the correct answer from the given options:

Random errors can be eliminated by:

- (a) taking number of observations and their mean. ✓
- (b) measuring the quantity with more than one instrument
- (c) eliminating the cause
- (d) careful observations

(i) Two projectiles are fired from the same point with the same speed at angles of projection 60° and 30° respectively. Which one of the following is true?

- (a) The range will be same ✓
- (b) Their maximum height will be same
- (c) Their landing velocity will be same
- (d) their time of flight will be same

(ii) The rate of change of linear momentum of a body is called a:

- (a) Linear force
- (b) Angular force
- (c) Power
- (d) Impulse ✓

(iii) If the external torque acting on a body is zero, then its

- (a) angular momentum is zero
- (b) angular momentum is conserved ✓
- (c) angular acceleration is maximum
- (d) rotational motion is maximum

(iv) When one joule of work is done on a body in one second, power of body is said to be:

- (a) One watt ✓
- (b) 0.5 watt
- (c) zero
- (d) 100 watt

(v) One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston:

- (a) increases by $2\Delta p$ ✓
- (b) increases by $\Delta p/2$

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- (c) increases by Δp
- (vii) If the cross-sectional area of a pipe decreases, what happens to the fluid velocity?
- (a) Increases ✓
(c) Remains the same
- (viii) A $2\mu\text{C}$ point charge is located at a distance "d" away from another point charge, what is the ratio of F_{12}/F_{21} ?
- (a) $1/3$ (b) 3 (c) 1 ✓ (d) 12
- (ix) Capacitors are used in electric power supply system to:
- (a) Improve power factor (b) Reduce line current
(c) Provide voltage stability ✓ (d) switching
- (x) A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will:
- (a) be quadrupled (b) be doubled ✓
(c) be halved (d) remain unchanged
- (xi) A simple harmonic oscillator has amplitude A and time period t . Its maximum speed is:
- (a) $\frac{4A}{t}$ (b) $\frac{2A}{t}$ (c) $\frac{4\pi A}{t}$ (d) $\frac{2\pi A}{t}$
- (xii) The speed of sound in a gas is proportional to:
- (a) square root of isothermal elasticity ✓
(b) square root of adiabatic elasticity
(c) isothermal elasticity (d) adiabatic elasticity
- (xiii) Optically active substances are those substances which:
- (a) produce polarized light
(b) rotate the plane of polarization of polarized light ✓
(c) produce double refraction
(d) convert a plane polarized light into circularly polarized light
- (xiv) Process of mapping the sampled analog voltage values to digital voltage levels is called:
- (a) sampling (b) sampling frequency
(c) quantizing ✓ (d) encoding
- (xv) The S.I unit of angular momentum is:
- (a) $\text{J} - \text{s}$ ✓ (b) $\frac{\text{J}}{\text{s}}$ (c) $\frac{\text{s}}{\text{J}}$ (d) $\frac{\text{J}}{\text{s}}$
- (xvi) The angle between centripetal and tangential acceleration in circular motion is:
- (a) 180° (b) 90° ✓ (c) 0° (d) 45°
- (xvii) Distance between two consecutive nodes in a standing wave is:

(a) $\frac{\lambda}{4}$

(b) λ

(c) $\frac{\lambda}{2}$ ✓

(d) $\frac{\lambda}{3}$

SECTION 'B' (Short-Answer Questions) (32)

NOTE: Answer any NINE part questions from this section. All questions carry equal marks. Draw diagrams where necessary.

2 (i) Find the dimensions of the following.

(a) work (b) energy (c) power (d) momentum

ANSWER:

(a) DIMENSIONS OF WORK

$$\text{Work} = \text{Force} \cdot \text{Displacement}$$

$$\text{Work} = F \cdot d$$

$$\text{Work} = m \cdot a \cdot d$$

We know that the dimension of mass is M , the dimension of acceleration is LT^{-2} , and the dimension of distance is L . Therefore

$$\text{Work} = [M][LT^{-2}][L]$$

$$\text{Work} = [ML^2T^{-2}]$$

(b) DIMENSIONS OF ENERGY

$$\text{Energy} = \text{Mass} \cdot (\text{Speed of light})^2$$

$$\text{Energy} = mc^2$$

We know that the dimension of mass is M , and the dimension of speed of light is LT^{-1} . Therefore

$$\text{Energy} = [M][LT^{-1}]^2$$

$$\text{Energy} = [M][L^2T^{-2}]$$

$$\text{Energy} = [ML^2T^{-2}]$$

(c) DIMENSIONS OF POWER

$$\text{Power} = \frac{\text{Work}}{\text{Time}}$$

$$\text{Power} = \frac{F \cdot d}{T} = F \left(\frac{d}{T} \right)$$

$$\text{Power} = FV$$

$$\text{Power} = m a V$$

We know that the dimension of mass is M , the dimension of acceleration is LT^{-2} , and the dimension of velocity is LT^{-1} . Therefore

$$\text{Power} = [M][LT^{-2}][LT^{-1}]$$

$$\text{Power} = [MLT^{-2}LT^{-1}]$$

$$\text{Power} = [ML^2T^{-3}]$$

(d) DIMENSIONS OF MOMENTUM

$$\text{Momentum} = \text{Mass} \cdot \text{Velocity}$$

$$\text{Momentum} = m \cdot V$$

We know that the dimension of mass is M , and the dimension of velocity is LT^{-1} . Therefore

$$\text{Momentum} = [M][LT^{-1}]$$

$$\text{Momentum} = [MLT^{-1}]$$

2 (ii) Two tug boats are towing a ship each exerts a force of 6000 N, the angle between two ropes is 60° . Calculate the resultant force on the ship?

ANSWER:

DATA:

Force exerted by First tug boat = $F_1 = 6000\text{ N}$
 Force exerted by Second tug boat = $F_2 = 6000\text{ N}$
 Angle between ropes = $\theta = 60^\circ$
 Resultant of force = $F = ?$

SOLUTION:

$$\therefore F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$F = \sqrt{(6000)^2 + (6000)^2 + 2(6000)(6000)\cos(60^\circ)}$$

$$F = \sqrt{36000000 + 36000000 + 2(36000000)\left(\frac{1}{2}\right)}$$

$$F = \sqrt{72000000 + 36000000}$$

$$F = \sqrt{108000000}$$

$$F = 10392.304\text{ N}$$

2 (iii) State the law of conservation of momentum.

ANSWER:

LAW OF CONSERVATION OF MOMENTUM

It states that "In a closed system, the total momentum before a collision is equal to the total momentum after the collision".

Mathematically

Momentum before collision = Momentum after collision

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

2 (iv) (a) What is the angular momentum of a 2.9 kg uniform cylindrical grinding wheel of radius 20 cm when rotating 1550 rpm?

(b) How much torque is required to stop it in 6s?

ANSWER:

DATA:

Mass of the cylindrical grinding wheel = $m = 2.9\text{ kg}$

Radius of cylindrical grinding wheel = $r = 20\text{ cm} = \frac{20}{100} = 0.2\text{ m}$

Angular velocity = $\omega = 1550\text{ rpm} = \frac{1550}{60} \times 2\pi = 162.3\text{ rad/s}$

Time = $t = 6 \text{ sec.}$

Angular momentum = $L = ?$

Torque = $\tau = ?$

SOLUTION:

We can find the angular momentum of the cylindrical grinding wheel by using the moment of inertia of a Solid Cylinder

$$\therefore I = \frac{1}{2}mr^2$$

$$I = \frac{1}{2}(2.9)(0.2)^2$$

$$I = 1.45(0.04)$$

$$I = 0.058 \text{ kgm}^2$$

$$\therefore L = I\omega$$

$$L = (0.058)(162.3)$$

$$L = 9.414 \text{ kg.m}^2/\text{s}^2$$

Let L' be the final angular momentum of the cylinder which is equal to zero, then the relation between angular momentum and torque will be:

$$\therefore \tau = \frac{L' - L}{t}$$

$$\tau = \frac{0 - 9.414}{6}$$

$$\tau = -\frac{9.414}{6}$$

$$\tau = -1.6 \text{ N.m}$$

Q) What is the law of conservation of energy?

ANSWER:

LAW OF CONSERVATION OF ENERGY

State that:

In an isolated system, energy can neither be created nor be destroyed but it can only be transformed from one form to another form.

The total mechanical energy of the system remains constant if the forces acting on the system are conservative in nature. Potential energy and kinetic energies keep interchanging with each other.

In a hydraulic press a force of 20 N is applied to a piston of area 0.20 m^2 . The area of the other piston is 2.0 m^2 . What is (a) the pressure transmitted through the fluid; (b) the force on the piston?

ANSWER:

A:

Force applied on small piston = $F_1 = 20 \text{ N}$

Area of small piston = $A_1 = 0.20 \text{ m}^2$

Area of Large piston = $A_2 = 2.0 \text{ m}^2$

Pressure = $P = ?$

Force applied on Large piston = $F_2 = ?$

SOLUTION:

For Pressure:

According to Pascal's law, the pressure exerted on the fluid in the same is equally transmitted, therefore, the pressure in both cylinders is

$$P = \frac{F}{A}$$

$$P = \frac{0.20}{20}$$

$$P = 100 \text{ Pa}$$

For Force applied on large piston:

$$P = \frac{F_2}{A_2}$$

$$P \times A_2 = F_2$$

$$F_2 = 100 \times 2$$

$$F_2 = 200 \text{ N}$$

2 (vii) Describe terminal velocity in liquids

ANSWER:

TERMINAL VELOCITY IN LIQUIDS

Terminal velocity is defined as the highest velocity attained falling through a fluid. It is observed when the sum of buoyancy is equal to the downward gravity force acting on it. acceleration of the object is zero as the net force acting on the object is zero. In fluid mechanics, for an object to attain its terminal velocity, it must have a constant speed against the force exerted by the fluid it is moving.

The mathematical representation of terminal velocity is:

$$V_t = \frac{2gr^2}{9\eta} (p - \sigma)$$

2 (viii) An electron is liberated from the lower of two large parallel plates separated by a distance $h = 2 \text{ cm}$. The upper plate is at a potential of 2400 volts relative to the lower. How long does the electron take to reach it?

ANSWER:

DATA:

Distance between Parallel plates = $h = d = 2 \text{ cm} = 0.02 \text{ m}$

Potential Difference between plates = $\Delta V = 2400 \text{ V}$

Charge of Electron = $q_e = 1.6 \times 10^{-19} \text{ C}$

Mass of Electron = $M_e = 9.1 \times 10^{-31} \text{ kg}$

Time taken to reach = $t = ?$

SOLUTION:

$$\therefore t = \frac{F}{q} \quad E = \frac{\Delta V}{d}$$

$$\frac{F}{q} = \frac{\Delta V}{d}$$

$$F = ma$$

$$\frac{ma}{q} = \frac{\Delta V}{d}$$

$$a = \frac{q_e \Delta V}{md}$$

$$a = \frac{(1.6 \times 10^{-19})(2400)}{(9.1 \times 10^{-31})(0.02)}$$

$$a = \frac{3.84 \times 10^{-16}}{1.82 \times 10^{-32}}$$

$$a = 2.109 \times 10^{16} \text{ ms}^{-2}$$

$$\therefore S = v_i t + \frac{1}{2} at^2$$

$$d = (0)t + \frac{1}{2} at^2$$

$$d = \frac{1}{2} at^2$$

$$t^2 = \frac{2(0.02)}{2.109 \times 10^{16}}$$

$$t^2 = \frac{0.04}{2.109 \times 10^{16}}$$

$$t^2 = 1.895 \times 10^{-18}$$

Taking square root on both sides

$$t = 1.3763 \times 10^{-9} \approx 1.4 \times 10^{-9} \text{ seconds}$$

2 (ix) Drive expressions for the combined capacitance of two capacitors

(a) connected in series,

(b) connected in parallel.

ANSWER:**(a) WHEN TWO CAPACITORS CONNECTED IN SERIES**

Two capacitors are connected in series combination. When a potential V is applied across them, the same amount Q of charge will appear on each of the capacitor. The charges on capacitors connected in series are the same. i.e.

$$Q = Q_1 = Q_2$$

In general, the total potential difference across any number of capacitors connected in series is the sum of the potential differences across the individual capacitors. i.e.

$$V = V_1 + V_2$$

To find an expression for equivalent capacitance C_{eq} , find the potential difference across each actual capacitor and equivalent capacitor.

$$\because V = \frac{Q}{C}, \quad V_1 = \frac{Q}{C_1}, \quad \text{and}, \quad V_2 = \frac{Q}{C_2}$$

$$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{Q}{C} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

This expression shows that the reciprocal of the equivalent capacitance is the algebraic sum of the reciprocal of the individual capacitance. Thus equivalent capacitance of a series combination is always less than individual capacitance.

(b) WHEN TWO CAPACITORS CONNECTED IN PARALLEL

Consider two capacitors are connected in parallel combination. We know that potential difference across each capacitors connected in parallel are same and equal to the potential difference applied across the combination i.e.,

$$V = V_1 = V_2$$

In general, when a potential difference V is applied across several capacitors connected in parallel, that potential difference V is same across each capacitor. The total charge Q stored on the equivalent capacitor is the sum of the charges stored on all the capacitors.

To find an expression for equivalent capacitance C_{eq} , find the charges on each actual capacitor and equivalent capacitor.

$$\because Q = CV, \quad Q_1 = C_1V, \quad \text{and}, \quad Q_2 = C_2V$$

The total charge on the parallel combination is given as,

$$Q = Q_1 + Q_2$$

$$C_{eq}V = C_1V + C_2V$$

$$C_{eq}V = V(C_1 + C_2)$$

$$C_{eq} = C_1 + C_2$$

Therefore, the equivalent capacitance of a parallel combination of capacitors is the algebraic sum of the individual capacitances and is a greater value than any of the individual capacitances.

2 (x) The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.50Ω , what is the maximum current that can be drawn from the battery

ANSWER:

DATA:

$$\text{Emf of Battery} = \epsilon = 12V$$

Internal resistance = $R = 0.50\Omega$

Maximum Current = $I = ?$

SOLUTION:

$$\epsilon = IR$$

$$I = \frac{\epsilon}{R}$$

$$I = \frac{12}{0.50}$$

$$I = 24A$$

2. (xi) A wire is hanging from the top of a tower such that the top is not visible due to darkness. How do you calculate the height of tower?

ANSWER:

For determination of height of the tower, we will attach a body at lower end of wire, then displace it from its mean position. When we displace it from its mean position and release so, it will execute simple harmonic motion like pendulum. We will determine the time period of this pendulum (wire and body system). After determining of time period we will put this value in the following formula to get length of wire.

$$T = 2\pi \sqrt{\frac{L}{g}}$$

In this way, we can calculate height of the tower easily.

2. (xii) A train sounds its whistle while passing by a railroad crossing. An observer at the crossing measures a frequency of 219 Hz as the train approaches the crossing and a frequency of 184 Hz as the train leaves. The speed of the sound is $340ms^{-1}$. Find the speed of the train and frequency of its whistle.

ANSWER:

DATA:

Frequency measures by observer = $f' = 219Hz$

Frequency of train leaves = $f'' = 184Hz$

Speed of the sound = $v = 340ms^{-1}$

Frequency of whistle = $f_s = ?$

Speed of train = $v_s = ?$

SOLUTION:

When train approaches to observer

$$f' = \left(\frac{v}{v - v_s} \right) f_s$$

$$f_s = \left(\frac{v - v_s}{v} \right) f'$$

$$f_s = \left(\frac{340 - v_s}{340} \right) (219) \quad \text{--- (1)}$$

When train moves away to observer

$$f'' = \left(\frac{v}{v + v_s} \right) f_s$$

$$f_s = \left(\frac{v + v_s}{v} \right) f''$$

$$f_s = \left(\frac{340 + v_s}{340} \right) (184) \quad \text{--- (ii)}$$

Since the actual frequency of the source is the same, therefore on equating equation (i) and (ii), we get

$$\left(\frac{340 - v_s}{340} \right) (219) = \left(\frac{340 + v_s}{340} \right) (184)$$

$$(340 - v_s)(219) = (340 + v_s)(184)$$

$$74460 - 219v_s = 62560 + 184v_s$$

$$74460 - 62560 = 184v_s + 219v_s$$

$$11900 = 403v_s$$

$$v_s = \frac{11900}{403}$$

$$v_s = 29.528 \text{ ms}^{-1}$$

On putting the value of " v_s ", we get

$$f_s = \left(\frac{340 - 29.528}{340} \right) (219)$$

$$f_s = \left(\frac{310.47}{340} \right) (219)$$

$$f_s = (0.9131)(219)$$

$$f_s = 199.98 \approx 200 \text{ Hz}$$

2 (xiii) Explain why two waves of significantly different frequencies cannot be coherent?

ANSWER: When two waves have significantly different frequencies, they cannot be coherent because their crest and troughs will not align in time; they will not have a constant phase relationship with each other.

2 (xiv) Write advantages and disadvantages of frequency modulation.

ANSWER:

ADVANTAGES OF FREQUENCY MODULATION

BETTER SOUND QUALITY:

FM provides better sound quality compared to AM, making it ideal for broadcasting music and high-fidelity audio.

NOISE IMMUNITY:

FM is less susceptible to noise and static interference, resulting in clear and more consistent reception.

WIDER FREQUENCY RANGE:

FM has a wider frequency range, allowing for more channels and better transmission of stereo signals.

HIGHER SIGNAL-TO-NOISE RATIO:

FM provides a higher signal-to-noise ratio, enhancing the overall signal quality and reducing the impact of background noise.

DISADVANTAGES OF FREQUENCY MODULATION**IMPROVED SIGNAL-TO-NOISE RATIO:**

Frequency modulation (FM) is less susceptible to noise and interference than other types of modulation, resulting in a clearer and more reliable signal.

ENHANCED AUDIO QUALITY:

FM is often used for audio signals, such as in radio broadcasting, and is known for its superior sound quality compared to other methods of modulation.

GREATER DYNAMIC RANGE:

FM allows for a wider range of signal strength, or dynamic range, allowing for more nuance and expression in the transmitted signal.

MORE EFFICIENT USE OF BANDWIDTH:

FM uses a wider frequency range than other methods of modulation, allowing for more information to be transmitted within a given bandwidth.

SECTION "C" (Detailed Answer Questions)(32)

NOTE: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

1. (a) What are the uses of dimensional analysis? Explain each of them.

ANSWER:**USES OF DIMENSIONAL ANALYSIS**

Dimensionality or dimensional analysis is a technique used in physics to check the consistency of equations and measurements. It involves analyzing the dimensions of each term in an equation to ensure that the units on both sides of the equation are equivalent. If the units on both sides do not match, then the equation is either incorrect or incomplete. Dimensional analysis can also be used to derive equations for physical quantities by analyzing the dimensions of the various terms involved.

For example, $S = V_i t + \frac{1}{2} a t^2$ expresses a relation between distance, speed, acceleration and time. The dimensions of each quantity are added on both sides and then found equal i.e., $[L]$. So, equation is homogenous equation or dimensionally correct.

2. (b) State the law of conservation of momentum. Provide an example from everyday life that demonstrates the principle of momentum conservation.

ANSWER:**LAW OF CONSERVATION OF MOMENTUM**

The law of conservation of linear momentum states that "if no external forces act on the system of two colliding objects, then the vector sum of the

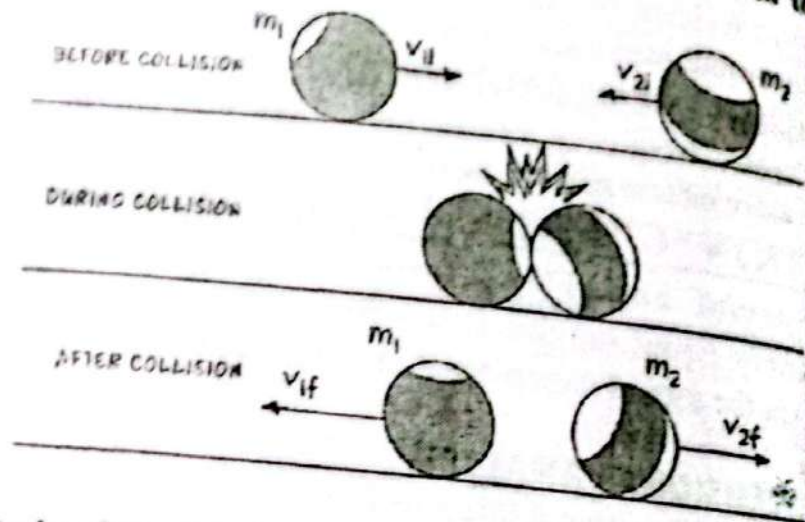
linear momentum of each body remains constant and is not affected by mutual interaction".

OR

The law of conservation of momentum states that if net external force on a system is zero, the total momentum of the system remains constant.

EXAMPLE OF LAW OF CONSERVATION OF MOMENTUM

One can easily understand the law of conservation of momentum by observing the complete process of collision of two objects moving with respective velocities towards each other. For this purpose, suppose two objects, say two balls having masses m_1 and m_2 move towards each other with velocities v_{1i} and v_{2i} respectively. The two objects or balls tend to collide with each other when the distance between them is nearly equal to zero. As a result of the collision, both the balls suffer the impact of equal and opposite forces in opposite directions that cause the balls to move in the opposite directions.



The magnitude of the forces acting on individual balls can be calculated by evaluating the rate of change in the momentum of the balls. It is observed theoretically as well as in practice that the sum of the momenta of both the balls before the collision is equal to the sum of the momenta of both the balls after the collision. The impact force generated during collision increases with an increase in the mass and velocity of the objects. This means that objects moving with a greater velocity or the objects with a significantly more mass value can cause severe destruction or produce a large magnitude impact force as such objects eventually possess a high momentum.

4. (a) Define work by variable force. Calculate the work done from a force-displacement graph.

ANSWER:

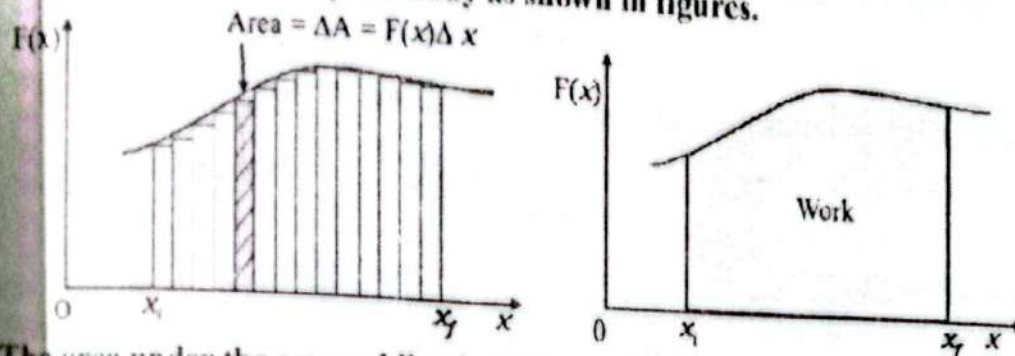
WORK DONE BY VARIABLE FORCE

Work done by variable force is a concept in physics that involves calculating the amount of energy transferred when an object is subjected to a variable force.

changes in magnitude or direction as the object moves. To calculate the work done by a variable force, you typically need to use calculus, specifically integration, because the force is not constant over the entire distance the object moves.

CALCULATION OF WORK DONE BY FORCE-DISPLACEMENT GRAPH

Consider a body covers displacement from x_i to x_f , when a variable force acts on it. To clarify the situation, we plot the graph between force and displacement covered by the body as shown in figures.



The area under the covered line is representing the work done by the body. To calculate the work done by the body we divide the covered displacement into small segments as

$$\Delta x_1, \Delta x_2, \Delta x_3 \dots \dots \Delta x_n$$

And the corresponding forces for each segment are

$$\vec{F}_{x1}, \vec{F}_{x2}, \vec{F}_{x3} \dots \dots \vec{F}_{xn}$$

We know the work done for each segment will be

$$W_1, W_2, W_3, \dots \dots W_n$$

The total work done in this case will be

$$W_T = W_1 + W_2 + W_3 + \dots \dots W_n$$

$$\because W = F \cdot x$$

$$W_T = \vec{F}_{x1}\Delta x_1 + \vec{F}_{x2}\Delta x_2 + \vec{F}_{x3}\Delta x_3 + \dots \dots \vec{F}_{xn}\Delta x_n$$

$$W_T = \sum_{i=1}^n \vec{F}_{xi}\Delta x_i$$

$$W_T = \vec{F}_i \cos \theta \Delta x_i$$

4. (b) State and explain Coulombs law. Apply it to calculate the electric field due to an isolated point charge.

ANSWER:

COULOMBS LAW

According to Coulomb's law "The force of attraction or repulsion between two charged things is directly proportional to the product of their charges and inversely proportional to the square of the distance between them".

OR

The electrostatic force F is directly proportional to the product of the magnitudes of the charges q_1 and q_2 and inversely proportional to the square of the distance r between the charges.

$$\vec{F} \propto q_1 q_2$$

$$\vec{F} \propto \frac{1}{r^2}$$

This result is known as the inverse square law.

$$\vec{F} = k \frac{q_1 q_2}{r^2}$$

Where k is known as the Coulomb's constant and $k = \frac{1}{4\pi\epsilon_0}$

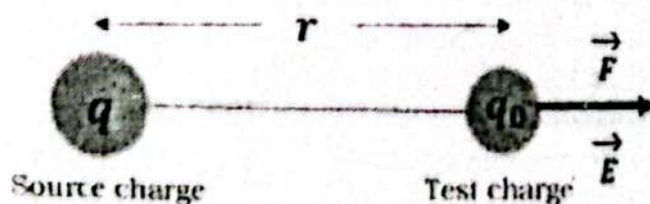
$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

EXPLANATION:

If two charged particles are brought near each other, they exert electrostatic force on each other. The direction of the force vectors depends on the signs of the charges. If the particles have the same sign of charge, they repel each other. That means that the force vector on each is directed away from the other particle. Beside this, If the particles have opposite sign of charge, they attract each other and the force vector on each is directed towards the other particle. The force of attraction and repulsion is dependent on the medium between the charges.

ELECTRIC FIELD DUE TO AN ISOLATED POINT CHARGE

To find the electric field due to a charged particle, consider a point charge q as a source charge. This charge creates an electric field at all points in the space surrounding it. A test charge q_0 is placed at a distance r from the source charge.



The test charge experiences electric force from which we can determine the direction of the electric field. According to Coulomb's law, the force exerted by q on the test charge is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^2} \hat{r}$$

where $\hat{r}$ is a unit vector directed from q toward q_0 and the direction of $\vec{F}$ is directly away from the source charge q . To calculate the electric field at a distance r from the point charges, we use

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{q_0} \vec{F}$$

$$\vec{E} = \frac{1}{q_0} \left(\frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r} \right)$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

The direction of electric field directly away from the point charge q and the magnitude of electric field at any given distance r can be expressed as:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\vec{E} = \left(\frac{q}{4\pi\epsilon_0} \right) \frac{1}{r^2}$$

$$\vec{E} = (\text{constant}) \frac{1}{r^2}$$

$$\vec{E} \propto \frac{1}{r^2}$$

This equation shows that the intensity of the electric field decreases with distance from the source charge(s). The field intensity is inversely proportional to the square of the distance from the charge(s). If there are n number of charged particles $q_1, q_2, q_3, \dots, q_n$, then the net electric force at these charges at a point. According to superposition principle,

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots \vec{F}_n$$

The net electric field of these charged particles

$$\vec{E} = \frac{\vec{F}_1}{q_0} + \frac{\vec{F}_2}{q_0} + \frac{\vec{F}_3}{q_0} + \dots \frac{\vec{F}_n}{q_0}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots \vec{E}_n$$

Therefore, the electric fields of these charges experienced by the test charge q_0 is vector sum of electric field of individual charge.

(a) Define simple harmonic motion (SHM). Discuss the key characteristics of a system undergoing SHM.

ANSWER:

SIMPLE HARMONIC MOTION

Simple harmonic motion is periodic motion that occurs whenever the restoring force is proportional to the displacement from equilibrium.

OR

Simple Harmonic Motion (SHM) is a type of oscillation or vibratory motion produced under the action of restoring force.

CHARACTERISTICS OF SIMPLE HARMONIC MOTION

The restoring force shall be directly proportional to the displacement from equilibrium position.

The restoring force shall be proportional to the inertia (elastic limit) of the system executing SHM.

3. The displacement from the equilibrium point on either side should be small.
4. Acceleration of the oscillating object should be proportional to the displacement ($a \propto -x$), the negative sign indicates that acceleration is directed towards the equilibrium position O.
5. (b) Define standing waves and explain how they are formed.

ANSWER:

STANDING WAVE

Standing Waves are the combination of two waves having same amplitude and frequency but moving in opposite directions in a string."

FORMATION OF STANDING WAVES

Stationary (Standing) waves occur when a string is tightly stretched between two rigid supports. If the string is plucked from half of its length, it extends the whole distance between the supports. This distance is equal to the wavelength of the transverse wave produced in the stretched string. A wave is reflected at the boundary and the reflected wave interferes with the incident wave so that the wave appears to stand still. Suppose an incident harmonic wave on a string, coming from left, hits a boundary where the string is fixed. The equation of incident harmonic sine wave is;

$$y_1 = A \sin(kx - \omega t)$$

and the equation of the reflected sine wave travelling to right is;

$$y_2 = A \sin(kx + \omega t)$$

y_1 and y_2 represent the displacements of incident and reflected waves. Applying the principle of superposition the motion of the stretched string is described by

$$y = y_1 + y_2 = A[\sin(kx - \omega t) + \sin(kx + \omega t)]$$

Using the trigonometric identity

$$\sin \alpha + \sin \beta = 2 \left[\frac{1}{2} \sin(\alpha + \beta) \right] \left[\frac{1}{2} \cos(\alpha - \beta) \right]$$

Here,

$$\alpha = kx - \omega t \quad \text{and} \quad \beta = kx + \omega t$$

We have;

$$y = 2A \left[\frac{1}{2} \sin\{(kx - \omega t) + (kx + \omega t)\} \right] \left[\frac{1}{2} \cos\{(kx - \omega t) - (kx + \omega t)\} \right]$$

$$y = 2A \left[\frac{1}{2} \sin(2kx) \right] \left[\frac{1}{2} \cos(-2\omega t) \right]$$

$$y = 2A(\sin kx)(\cos \omega t)$$

$$y = 2A \sin kx \cos \omega t$$

The resultant expression is the wave function of a standing wave obtained due to the superposition of two waves of the same amplitude and frequency moving with same speed in opposite direction along same axis.

YOUSUIFS MODEL QUESTION PAPER 2

PHYSICS (THEORY)

CLASS XI (SCIENCE)

Total Time: 3 Hours

Total Marks: 85

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

NOTE: This Section consists of 17 MCQs and all are to be answered.
Each MCQ carry 1 marks.

1. Choose the correct answer from the given options:

(i) Dimensions of kinetic energy is the same as that of:

(a) Acceleration (b) Velocity (c) Work ✓ (d) Force

(ii) At the top of a trajectory of a projectile, the acceleration is:

(a) maximum (b) minimum (c) zero (d) g ✓

(iii) A 3 kg bowling ball experiences a net force of 15 N . What will be its acceleration.

(a) 35 m/s^2 (b) 7 m/s^2 (c) 5 m/s^2 ✓ (d) 35 m/s^2

(iv) A stone of 2 kg is tied to a 0.50 m long string and swung around a circle at angular velocity of rad/s . The net torque on the stone about the center of the circle is

(a) 0 N.m (b) 6 N.m (c) 12 N.m (d) 72 N.m ✓

(v) Two bodies of mass 1 kg and 2 kg have equal momentum. Then the ratio of their kinetic energies is:

(a) $2 : 1$ ✓ (b) $3 : 1$ (c) $1 : 3$ (d) $1 : 1$

(vi) A rock suspended by a weighing scale weighs 5 N out of water and 3 N when submerged in water. What is the buoyant force on the rock?

(a) 3 N (b) 5 N ✓ (c) 8 N (d) 15 N

(vii) A fluid is undergoing "incompressible" flow. This means that:

(a) the pressure at a given point cannot change with time

(b) the velocity at a given point cannot change with time

(c) the velocity must be the same everywhere

(d) the density cannot change with time or location ✓

(viii) 12 J of work is to be done against an existence electric field to take a charge of 0.01 C from one-point A to another point B . The potential difference between B and A is

(a) 120 V

(b) 1200 V ✓

(c) 1.2 V

(ix) Energy stored in the capacitor is:

(a) $E = \frac{1}{4} CV$ (b) $E = \frac{1}{2} CV^2$ ✓(c) $E = CV^2$ (d) $E = \frac{1}{2} CV$

(x) What is a potentiometer primarily used for?

(a) Measuring electric current

(b) Measuring electric voltage ✓

(c) Measuring potential difference (voltage) ✓

(d) Measuring electric resistance

(xi) The total energy of a particle executing simple harmonic motion is proportional to:

(a) frequency of oscillation

(b) maximum velocity of motion

(c) amplitude of motion

(d) square of amplitude of motion ✓

(xii) The speed of a wave in a medium is 760 m/s. If 3600 waves pass through point in the medium in 2 minutes, then its wavelength is:

(a) 13.85 m

(b) 25.3 m ✓

(c) 41.5 m

(d) 57.2 m

(xiii) If the wavelength of an electromagnetic wave is about the size of a cricket ball, what type of radiation is it.

(a) X-ray

(c) Radio waves ✓

(b) Ultraviolet

(d) Visible light

(xiv) The process of superimposing signal frequency on the carrier wave is known as:

(a) transmission

(c) reception

(b) detection

(d) modulation ✓

(xv) If the velocity of a body is doubled and the mass is reduced to one-fourth, the kinetic energy will be:

(a) doubled

(b) unchanged ✓

(c) halved

(d) four times

(xvi) A body pushes a toy car, on a horizontal floor with a force of 20 N up to a displacement of 2 m, then work done by the gravity on the car is:

(a) 20 J ✓

(b) 10 J

(c) 5 J

(d) 0 J

(xvii) The experimental evidence of transverse nature of light is:

(a) diffraction

(c) polarization ✓

(b) interference

(d) dispersion

SECTION 'B' (Short-Answer Questions) (3)**NOTE:** Answer any NINE part questions from this section. Each question carries equal marks. Draw diagrams where necessary.

2 (i) The diameter of Moon is 3480 km. What is the volume of the Moon? How many Moons would be needed to create a volume equal to that of Earth? (Hint: Radius of Earth = 6380 km)

ANSWER:

DATA:

$$\text{Diameter of Moon} = D_m = 3480 \text{ km} = 3480 \times 1000 = 3.48 \times 10^6 \text{ m}$$

$$\text{Radius of Moon} = R_m = \frac{D_m}{2} = \frac{3.48 \times 10^6}{2} = 1.74 \times 10^6 \text{ m}$$

$$\text{Volume of the Moon} = V_m = ?$$

$$\text{Radius of Earth} = R_E = 6380 \text{ km} = 6380 \times 1000 = 6.38 \times 10^6 \text{ m}$$

$$\text{No. of Moon} = ?$$

SOLUTION:

Since the Moon is nearly spherical, so

$$V = \frac{4}{3} \pi r^3$$

$$V_m = \frac{4}{3} \pi (1.74 \times 10^6)^3$$

$$V_m = \frac{4}{3} \pi (5.268 \times 10^{18})$$

$$V_m = 2.21 \times 10^{19} \text{ m}^3$$

Since the Earth is nearly spherical, so

$$V_E = \frac{4}{3} \pi (6.38 \times 10^6)^3$$

$$V_E = \frac{4}{3} \pi (2.596 \times 10^{20})$$

$$V_E = 1.09 \times 10^{19} \text{ m}^3$$

$$\text{No. of Moon} = \frac{1.09 \times 10^{19}}{2.21 \times 10^{19} \text{ m}^3}$$

$$\text{No. of Moon} = 49.32$$

2 (ii) Can an object have an initial velocity of zero and still experience uniformly accelerated motion

ANSWER: Yes, it happens when an object or ball is dropped from a certain height, so at that moment its initial velocity is zero but it experiences a uniform acceleration because of acceleration due to gravity which is approximately equal to 9.8 ms^{-2} .

2 (iii) A helicopter weighs 3920 N. Calculate the force on it if it is ascending up at the rate of 2 m/s^2 . What will be the force on helicopter if it is moving up with a constant speed of 4 m/s ?

ANSWER:

DATA:

$$\text{Weight of the helicopter} = W = 3920 \text{ N}$$

$$\text{Mass of the car} = \frac{W}{g} = \frac{3920}{9.8} = 400 \text{ kg}$$

$$\text{Acceleration} = a = 2 \text{ ms}^{-2}$$

$$\text{Force} = F_1 = ?$$

$$\text{Constant speed} = V = 4 \text{ ms}^{-1}$$

$$\text{Force at constant speed} = F_2 = ?$$

SOLUTION:

CASE 1: Since the helicopter is ascending, therefore the net force acting on it will be

$$\therefore F - W = ma$$

$$F = ma + W$$

$$F = (400)(2) + 3920$$

$$F = 800 + 3920$$

$$F = 4720 \text{ N}$$

CASE 2: Since the helicopter is ascending with constant speed, therefore the net force acting on it will be

$$\therefore F - W = ma$$

$$F = ma + W$$

Since the speed is constant, so $a = 0$

$$F = (400)(0) + 3920$$

$$F = 0 + 3920$$

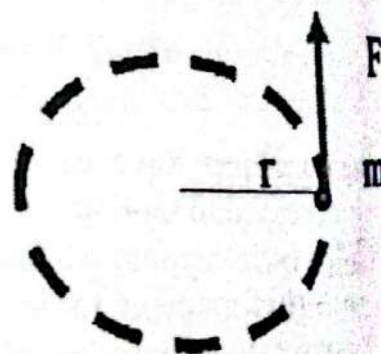
$$F = 3920 \text{ N}$$

2 (iv) Derive relationship between torque and angular acceleration.

ANSWER:

DERIVATION:

Consider a particle of mass m rotating in a circle of radius r at the end of a string whose mass is negligible as compared to mass of string. Assume that a single force F acts on mass as shown in the figure. The torque gives rise to the angular acceleration is



$$\tau = r F$$

If $F_{\tan}$ is the tangential component of the force and $a_{\tan}$ is the tangential component acceleration, then

According to Newton's second law of motion,

$$F_{\tan} = ma_{\tan} \text{ --- (i)}$$

We know that the tangential acceleration ($a_{\tan}$) is the product of angular acceleration and the distance from the axis of rotation to the point of application force i.e.

$$a_{\tan} = r\alpha$$

On substituting the value of tangential acceleration in equation (i), we get

$$F_{\tan} = m(r\alpha)$$

$$F_{\tan} = mra$$

If we multiply both sides by r , the equation becomes

$$F_{\tan} \times r = mra \times r$$

$$F_{\tan} \times r = mr^2\alpha$$

Since the product of force ($F_{\tan}$) acting on the body and the distance (r) is known as torque (τ), i.e.

$$\tau = mr^2\alpha$$

Since mr^2 is the moment of inertia (I) of a body

$$\tau = I\alpha$$

If the moment of inertia (I) is constant then the above equation will become

$$\tau \propto \alpha \text{ Derived}$$

2 (v) A man pulls a trolley through a distance of 10 m by applying a force of 50 N which makes an angle of 60° with the horizontal. Calculate the work done by the man?

ANSWER:

DATA:

$$\text{Distance} = s = 10\text{m}$$

$$\text{Force} = F = 50\text{N}$$

$$\text{angle} = \theta = 60^\circ$$

$$\text{work done} = W = ?$$

SOLUTION:

$$\therefore W = F \cdot s \cos\theta$$

$$W = (50)(10) \cos 60^\circ$$

$$W = (500)(0.5)$$

$$W = 250 \text{ J}$$

2 (vi) State Pascal's principle. Describe its two applications.

ANSWER:

PASCAL'S PRINCIPLE

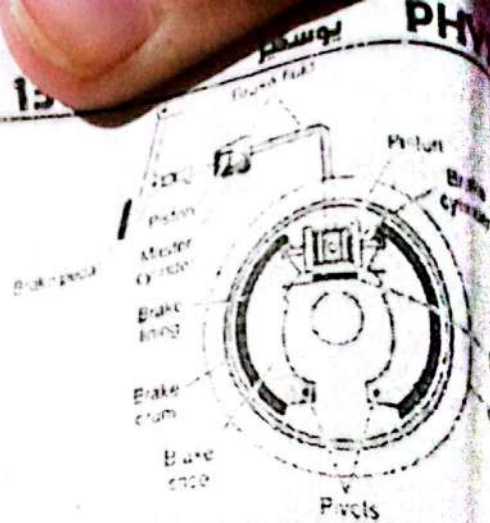
Pascal's principle states that "when a change in pressure is applied at a point to any static fluid, it is transmitted perfectly to all portions of the fluid and to the wall of the container".

APPLICATIONS

1. AUTOMOBILE HYDRAULIC BRAKE SYSTEM:

MODEL PAPERS XI (S)

The rear wheel hydraulic brake system of a front-wheel-drive is an application of Pascal's principle. When driver pushes the brake pedal, "The pressure on the piston in master cylinder is transmitted through the brake fluid to the two pistons in the brake cylinder."

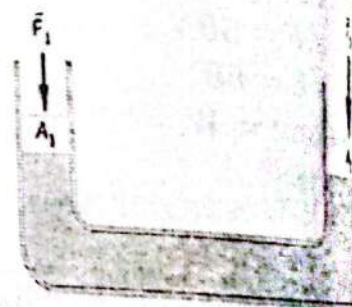


This transmitted pressure then forces the brake-cylinder pistons to push the brake shoes against the brake drum and stop the automobile. Now releasing the brake pedal releases the pressure on the pistons in the cylinder. The spring pulls the brake shoes away from the brake drum which allows the wheel to turn freely again.

2. HYDRAULIC LIFT OR HYDRAULIC JACK:

Hydraulic lift is able to raise up large weight up to relatively short distance.

The hydraulic lift, or jack is applications of hydraulics being used as a simple machine to multiply force. It contains an incompressible fluid in a U-shaped tube which is narrower at start and becomes wider area at end which is fixed with a movable piston on each side.



If a small force F_1 is applied to the small piston of the hydraulic lift shown in figure, the pressure is transmitted with in all directions. The pressure on the large piston is the same as the pressure on the small piston however, the force F_2 on the large piston is greater because of its large surface area which used to uplift the car.

2 (vii) Two spherical raindrops of equal size are falling through air with velocity of 0.08 m/s . If the drops join together forming a large spherical drop, what will be the new terminal velocity?

ANSWER:

DATA:

terminal velocity small raindrops = $V_t = 0.08 \text{ ms}^{-1}$

No. of drops = $N = 2$

terminal velocity large raindrops = $V_T = ?$

SOLUTION:

Since the terminal velocity of a single drop is

$$V_t = \frac{2gr^2}{9\eta}(\rho - \sigma)$$

Volume of large spherical drop = N (Volume of Small drops)

$$\frac{4}{3}\pi R^3 = 2\left(\frac{4}{3}\pi r^3\right)$$

$$\frac{4}{3}\pi R^3 = 2\left(\frac{4}{3}\pi r^3\right)$$

$$R^3 = 2r^3$$

taking cube root on both sides, we get

$$R = 2^{\frac{1}{3}}r$$

Let V_T be the terminal velocity of a large drop then

$$V_T = \frac{2gR^2}{9\eta}(\rho - \sigma)$$

$$V_T = \frac{2g\left(2^{\frac{1}{3}}r\right)^2}{9\eta}(\rho - \sigma)$$

$$V_T = \frac{2g \cdot 2^{\frac{2}{3}}r^2}{9\eta}(\rho - \sigma)$$

dividing corresponding sides of V_T by V_t , we get

$$\frac{V_T}{V_t} = \frac{\frac{2g \cdot 2^{\frac{2}{3}}r^2}{9\eta}(\rho - \sigma)}{\frac{2gr^2}{9\eta}(\rho - \sigma)}$$

$$\frac{V_T}{V_t} = \frac{\frac{2gr^2 \times 2^{\frac{2}{3}}}{9\eta}(\rho - \sigma)}{\frac{2gr^2}{9\eta}(\rho - \sigma)}$$

$$\frac{V_T}{V_t} = 2^{\frac{2}{3}}$$

$$V_T = 2^{\frac{2}{3}}V_t$$

$$V_T = 2^{\frac{2}{3}}(0.08)$$

$$V_T = 0.12699 \approx 0.13 \text{ ms}^{-1}$$

$$V_T = 0.13 \text{ ms}^{-1}$$

vi) Why do most objects tend to contain nearly equal numbers of positive and negative charges?

ANSWER: Every object in the universe tries to attain minimum energy. Carrying any charge means the existence of a force that would require energy, therefore most of the objects tend to contain nearly equal numbers

of positive and negative charges so the net charge on the object will be zero. Thus, there will be no non-zero forces and hence energy is minimized.

2 (ix) The capacitance of air-filled parallel-plate capacitor is 1.3 pF . If separation of the plates is doubled and wax is inserted between them, the capacitance is 2.6 pF . Find the dielectric constant of the wax.

ANSWER:

DATA:

Capacitance of air – filled capacitor $= C_{\text{Air}} = 1.3 \text{ pF}$

separation of the plates in air – filled capacitor $= d_{\text{Air}}$

Capacitance of wax filled capacitor $= C_{\text{Wax}} = 2.6 \text{ pF}$

Separation of the plates in wax filled capacitor $= d_{\text{Wax}} = 2d_{\text{Air}}$

Dielectric constant of the air $= \epsilon_r = 1.0006 \approx 1$

Dielectric constant of the wax $= \epsilon_r' = ?$

SOLUTION:

For Air

$$\therefore C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C = \frac{\epsilon_0 (1) A}{d_{\text{Air}}}$$

$$d_{\text{Air}} = \frac{\epsilon_0 A}{C}$$

For wax

$$\therefore C_0 = \frac{\epsilon_0 \epsilon_r' A}{2d_{\text{Air}}}$$

$$\text{But } d_{\text{Air}} = \frac{\epsilon_0 A}{C}$$

$$C_0 = \frac{\epsilon_0 \epsilon_r' A}{2 \frac{\epsilon_0 A}{C}}$$

$$C_0 = \frac{\epsilon_r'}{2} \times C$$

$$\frac{2C_0}{C} = \epsilon_r'$$

$$\epsilon_r' = \frac{2(2.6)}{1.3}$$

$$\epsilon_r' = 4.0$$

2 (x) What is the difference between potential difference and emf?

ANSWER:

DIFFERENCE BETWEEN POTENTIAL DIFFERENCE AND EMF

ELECTROMOTIVE FORCE	POTENTIAL DIFFERENCE
It is the amount of energy supply to one coulomb of charge.	The amount of energy used by one coulomb of charge in moving from one point to another.
The electromotive force transfers energy throughout the circuit.	The potential difference is the portion of energy between any two points on a circuit.
The electromotive force can be induced in the magnetic, electric.	The potential difference can only be generated in an electric field.
It does not depend on circuit resistance.	It directly depends on the resistance between two points of measurement.
It is always greater than the voltage	It is always lesser than EMF

(xi) The period of oscillation of an object in an ideal spring and mass system is 0.50 s and the amplitude is 5.0 cm. what is the speed at the equilibrium point? and the acceleration at the point of maximum extension of the spring.

ANSWER:

DATA:

Period of oscillation = $T = 0.50s$

Amplitude = $y_0 = 5.0cm = \frac{5}{100} = 0.05m$

speed at the equilibrium point = $v_0 = ?$

Acceleration at the point of maximum extension = $a = ?$

SOLUTION:

$$\therefore T = \frac{2\pi}{\omega}$$

$$\omega = \frac{2\pi}{T}$$

$$\omega = \frac{6.283}{0.50}$$

$$\omega = 12.566$$

The speed of the object at the equilibrium point or mean position is maximum when body executing SHM, therefore

$$\therefore v_0 = \omega y$$

$$v_0 = (12.566)(0.05)$$

$$v_0 = 0.628 \text{ m s}^{-1}$$

$$v_0 = 0.628 \text{ m s}^{-1} \times 10^2$$

$$v_0 = 62.8 \text{ cm s}^{-1}$$

Acceleration at the extreme position

$$\therefore a = \omega^2 y$$

$$a = (12.566)^2 (0.05)$$

$$a = (157.91)(0.05)$$

$$a = 7.895 \text{ ms}^{-2}$$

2 (vii) What are the conditions for the interference of waves

ANSWER:

CONDITIONS FOR THE INTERFERENCE OF WAVES

Following are the conditions for the interference of sound waves

- (i) The two sources should be coherent i.e. they produce wave-trains of same frequency.
- (ii) The amplitude of the wave-trains must be equal.
- (iii) Two wave-trains produced from the source must travel along same direction and with the equal speed.
- (iv) They must overlap to superpose.

2 (viii) A monochromatic light of wavelength $6900 \text{ \AA}$ is used to illuminate two parallel slits. On a screen that is 3.30 m away from the slits, interference fringes are observed. The distance between adjacent bright fringes in the centre of the pattern is 1.80 cm . what is the distance between the slits.

ANSWER:

DATA:

Wavelength $\lambda = 6900 \text{ \AA} = \frac{6900}{10^{10}} = 6.9 \times 10^{-7} \text{ m}$

Distance between a screen and slits $= L = 3.30 \text{ m}$

distance between adjacent bright fringes $= \Delta x = 1.80 \text{ cm}$
 $= 0.018 \text{ m}$

Distance between slits $= d = ?$

SOLUTION:

$$\Delta x = \frac{L}{d} \lambda$$

$$d = \frac{L \lambda}{\Delta x} = \frac{3.30}{0.018} (9.6 \times 10^{-7})$$

$$d = (183.33)(9.6 \times 10^{-7})$$

$$d = 1.265 \times 10^{-4} \text{ m}$$

2 (xiv) What are limitations of amplitude modulation?

ANSWER:

LIMITATIONS OF AMPLITUDE MODULATION

- (i) This limits the application of amplitude modulation to radios. A system used for one to one communication only.
- (ii) This deteriorates the quality of the original signal at the receiver and causes problems in the quality of the signal.

- (iii) The efficiency of such a system is very low as it consumes a lot of power for modulations and this causes a considerable operational cost.
- (iv) The quality of the audio signal is poor.
- (v) This modulation will require more than twice the amplitude-frequency to modulate the signal with a carrier.
- (vi) Low efficiency as the information is contained only in the sideband.

SECTION "C" (Detailed Answer Questions)(32)

NOTE: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

3. (a) What is the significance of the area under a velocity-time graph in the context of accelerated motion?

ANSWER: The area under a velocity-time graph in the context of accelerated motion is very significant because it represents the displacement.

3. (b) Derive the formula for centripetal acceleration using fundamental principles and equations, illustrating each step of the derivation

ANSWER: (See Question No. 4(b), 2024 P.E. & P.M.)

4. (a) Discuss buoyancy in liquids and gases.

ANSWER:

BUOYANCY FORCE

The buoyant force acts in the opposite direction to gravity, resulting in an apparent reduction in the weight of the object when it is submerged in a fluid.

For an object fully submerged in a fluid, the volume of displaced fluid is equal to the volume of the object itself. If the object is partially submerged, only the submerged portion contributes to the volume of displaced fluid. The buoyant force is the reason why objects appear to be lighter when submerged in water or other fluids. It is also the principle behind the floating of ships and other vessels, as the buoyant force is greater than the weight of the ship, allowing it to stay afloat.

BUOYANCY IN LIQUIDS:

(i) If the weight of the submerged object is greater than buoyant force, the object sinks.

(ii) If the weight of object is equal to the buoyant force acting upwards on the submerged object, it remains at any level in fluid, like a fish.

(iii) If the buoyant force is greater than the weight of object which is completely submerged, it rises to the surface and floats.

For example, a ship that is launched into sea, sinks into the ocean until the weight of the water it displaces is just equal to its own weight. As the ship is loaded, it sinks deeper, displacing more water, and so the magnitude of the buoyant force continuously matches the weight of the ship and its cargo.

BUOYANCY IN GASES

We live at the bottom of ocean of air and look upward at balloons and lighter than air objects are drifting above us. Air pressure acting up against an object immersed in air is greater than the pressure above pushing down. The buoyancy is equal to the weight of fluid displaced. Archimedes' principle also implies to air which is stated as: 'An object surrounded by air is buoyed up by a force equal to the weight of the air displaced.'

A cubic meter of air at normal atmospheric pressure and room temperature has a mass of about 1.2 Kg, so its weight becomes around 12 N.

(i) If the mass of the 1 m^3 object is greater than 1.2 Kg, it falls to ground, when released.

(ii) If an object of this size has mass of less than 1.2 Kg, buoyant force is greater than weight and it rises in air.

Because gas (Hydrogen or Helium) filled in balloons that rises in air are less dense than air. Air becomes less dense at high altitude; a lesser weight of air is displaced per given volume as the balloon rises. When the weight of displaced air equals the total weight of the balloon, upward motion of balloon ceases. It can be stated as when the buoyant force on the balloon equals its weight, the balloon ceases to rise.

4. (b) Calculate combined capacitance of capacitors in series and in parallel.

ANSWER:**COMBINED CAPACITANCE OF A SERIES COMBINATION OF CAPACITORS**

Two or more capacitors are connected in series combination. When potential V is applied across them, the same amount Q of charge will appear on each of the capacitor. The charges on capacitors connected in series are the same. i.e.

$$Q = Q_1 = Q_2 = Q_3$$

In general, the total potential difference across any number of capacitors connected in series is the sum of the potential differences across individual capacitors. i.e.

$$V = V_1 + V_2 + V_3 + \dots \dots V_n$$

To find an expression for equivalent capacitance C_{eq} , find the potential difference across each actual capacitor and equivalent capacitor.

$$\therefore V = \frac{Q}{C}, \quad V_1 = \frac{Q}{C_1}, \quad V_2 = \frac{Q}{C_2}, \quad \text{and}, \quad V_3 = \frac{Q}{C_3}$$

$$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} + \dots \dots \frac{Q}{C_n}$$

$$\frac{Q}{C} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \dots \frac{1}{C_n} \right)$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \dots \frac{1}{C_n}$$

expression shows that the reciprocal of the equivalent capacitance is the algebraic sum of the reciprocal of the individual capacitance. Thus, the equivalent capacitance of a series combination is always less than any individual capacitance.

COMBINED CAPACITANCE OF A PARALLEL COMBINATION OF CAPACITORS

Consider two or capacitors are connected in parallel combination. We know that potential difference across each capacitors connected in parallel is the same and equal to the potential difference applied across the combination. i.e.,

$$V = V_1 = V_2 = V_3$$

In general, when a potential difference V is applied across several capacitors connected in parallel, that potential difference V is same across each capacitor. The total charge Q stored on the equivalent capacitor is the sum of the charges stored on all the capacitors.

To find an expression for equivalent capacitance C_{eq} , find the charge on each actual capacitor and equivalent capacitor.

$\therefore Q = CV$, $Q_1 = C_1V$, $Q_2 = C_2V$, and, $Q_3 = C_3V$
The total charge on the parallel combination is given as

$$\begin{aligned} Q &= Q_1 + Q_2 + Q_3 + \dots \dots Q_n \\ C_{eq}V &= C_1V + C_2V + C_3V + \dots \dots + C_nV \\ C_{eq}V &= V(C_1 + C_2 + C_3 + \dots \dots + C_n) \\ C_{eq} &= C_1 + C_2 + C_3 + \dots \dots + C_n \end{aligned}$$

Therefore, the equivalent capacitance of a parallel combination of capacitors is the algebraic sum of the individual capacitances and has greater value than any of the individual capacitances.

(a) Define resistivity. Explain dependence of resistivity on temperature.

ANSWER:

RESISTIVITY

Resistivity is the electrical resistance of a conductor of unit cross-sectional area and unit length. It is a characteristic property of the material and useful in comparing various materials on the basis of their ability to conduct electric currents.

DEPENDENCE OF RESISTIVITY ON TEMPERATURE:

Resistivity depend on the on temperature. At a higher temperature, the thermal energy is greater; the ions vibrate with larger amplitudes. As a result, the electrons collide more frequently with the ions. With less time to accelerate between collisions, they acquire a smaller drift speed, thus, the current is smaller for a given electric field. Therefore, as the

temperature of a metal is raised, its resistivity increases. The temperature of a glowing incandescent light bulb reaches a temperature of about 2500°C. Its resistance is significantly higher than at room temperature. In semiconducting materials, the relation between resistivity and temperature is more complex. In fairly wide range of temperatures (about 500°C),

$$\Delta R \propto R_0$$

$$\Delta R \propto \Delta T$$

$$R_t = R_0 (1 + \alpha \Delta T)$$

$$\rho_t = \rho_0 (1 + \alpha \Delta T)$$

Where, ρ_t = resistivity at temperature $T^\circ\text{C}$

ρ_0 = resistivity at temperature $T^\circ\text{C}$

α = linear temperature coefficient of resistivity and has SI unit $^\circ\text{C}^{-1}$. Note that for semiconductors, $\alpha < 0$. A negative temperature coefficient means that the resistivity decreases with increasing temperature, as for metals that are good conductors at room temperature. However, in semiconductors the number of free electrons (conduction electrons or holes) per unit volume increases with temperature; with more carriers, the resistivity is smaller.

5. (b) Discuss the conditions required for interference and provide examples of interference in daily life.

ANSWER:

CONDITIONS FOR INTERFERENCE OF LIGHT: COHERENT AND INCOHERENT SOURCES:

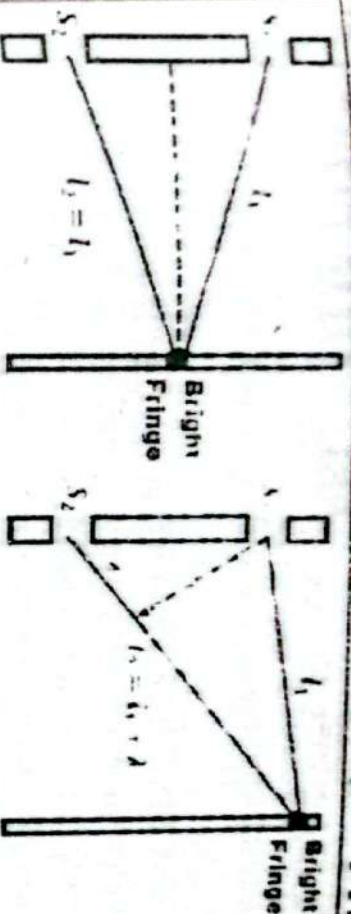
Although any number of waves can in principle interfere, but keep the case simple we consider here the interference of only two waves to produce observable interference effect, it is necessary to have

1. Two coherent sources. i.e., they must have the same frequency.
2. The two waves must be monochromatic i.e. they must have the same wavelength.
3. Be always in phase with each other or have a constant phase difference.
4. For Constructive Interference (Bright Fringes) the path difference must be an integral multiple of the wavelength.

$$\text{Path difference} = 2m\pi \text{ rad}$$

$$\text{Path difference } \Delta S = m\lambda$$

$$(m = 0, \pm 1, \pm 2, \pm 3, \dots)$$

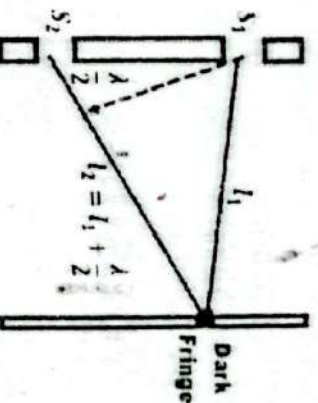


5. For Destructive Interference (Dark Fringes) the phase difference must be

Phase difference $= (m + \frac{1}{2}) 2\pi \text{ rad}$

Path difference $\Delta S = (m + \frac{1}{2}) \lambda$

($m = 0, \pm 1, \pm 2, \pm 3, \dots$)



DAILY LIFE EXAMPLES OF INTERFERENCE

SOAP BUBBLES

Soap bubbles are yet another example of the interference of light in real life. Soap bubbles are spherical in shape and have a thin transparent layer made up of soap solution. When the white light falls on the surface of the soap bubble and gets reflected from the top and bottom surface of the soap bubble, only a few colours contained by the white light tend to experience constructive interference, while the rest colours undergo destructive interference of light. The colours that encounter constructive interference are visible, while the rest colours get suppressed.

COLOUR PATCHES ON WET ROADS

One can easily observe colour patches formed on the roads during the rainy season. These colour patches on wet roads are basically formed due to the phenomenon of interference of white light. A thin layer of oil, deposited on the surface of the wet road, is primarily responsible for the constructive and destructive interference of the colours contained by the incident white light. When the light is made to strike the surface of the road, it gets reflected by the top and the bottom surface of the oil layer. The colours contained by the light that suffer a constructive interference are easily visible to the observer, while the colours that undergo destructive interference get hindered.

ANTI-REFLECTIVE COATING

The anti-reflective coating present on the surface of lenses of the spectacles, camera, etc. is one of the prime applications of interference of light in real life. The main purpose of such anti-reflective coatings is to reduce the intensity of the light falling on the surface of the object. A similar

interference phenomenon can also be observed on the surface of windows.

OIL ON THE SURFACE OF WATER

Oil and water have different chemical and physical properties. The refractive index of oil is comparatively less than the water. This is the reason why the two fluids do not get mixed with each other and oil tends to float on the surface of the water. When a ray of light hits the surface of the oil, it gets reflected by the top and the bottom surface of the layer. This causes the interference of light contained by the light radiation to undergo constructive and destructive interference. The construction interference of colours of light leads to the formation of a multi-coloured visible pattern on the surface of the oil.

YOUSUIFS MODEL QUESTION PAPER PHYSICS (THEORY) CLASS XI (SCIENCE)

Total Time: 3 Hours

Total Marks: 80

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

NOTE: This Section consists of 17 MCQs and all are to be answered. Each MCQ carry 1 marks.

1. Choose the correct answer from the given options:
 - (i) The respective number of significant figures for the numbers 23.0, 0.0003 and 2.1×10^{-3} are:

(a) 5, 1, 2 ✓	(b) 5, 1, 5	(c) 5, 5, 2	(d) 4, 4, 2
---------------	-------------	-------------	-------------
 - (ii) To get a resultant displacement of 10m, two displacement vectors of magnitude 6m and 8m should be combined:

(a) Parallel	(b) Antiparallel
(c) At an angle of 45°	(d) Perpendicular to each other ✓
 - (iii) The motion of rocket in the space is according to the law of conservation of:

(a) Energy	(b) linear momentum ✓
(c) mass	(d) angular momentum
 - (iv) An object rotating about a fixed axis, I is its rotational inertia and α is its angular acceleration. Its

(a) is the definition of torque ✓
(b) is the definition of rotational inertia
(c) is definition of angular acceleration
(d) follows directly from Newton's second law.

(v) A body of mass 5 kg is moving with a momentum of kg m/s . A force of 0.2 N acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy is:

- (a) 2.8 J (b) 3.2 J (c) 3.8 J (d) 4.4 J ✓

(vi) You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what happens to the water level?

- (a) the water overflows the rim
(b) the water level drops below the rim
(c) the water level stays at the rim ✓
(d) it depends on the difference in density between water and ice

(vii) Which of the following is associated with the law of conservation of energy in fluids?

- (a) Archimedes' principle (b) Bernoulli's principle ✓
(c) Pascal's principle (d) equation of continuity

(viii) The force between two charges placed in air is F . If air is replaced by a medium of relative permittivity ϵ_r , then force is reduced to:

- (a) $F\epsilon_r$ ✓ (b) F/ϵ_r (c) ϵ_r/F (d) $\epsilon\epsilon_r$

(ix) The capacitance C is charged through a resistor R . The time constant of the charging circuit is given by

- (a) C/R (b) $1/RC$ (c) RC ✓ (d) R/C

(x) Reciprocal of resistance is called:

- (a) conductance ✓ (b) resistivity
(c) resonance (d) capacitance

(xi) A simple harmonic oscillator with a natural frequency ω_N is forced to oscillate with a driving frequency ω_d . The Resonance occurs when:

- (a) $\omega_N < \omega_d$ (b) $\omega_N > \omega_d$
(c) $\omega_N = \omega_d$ (d) $\omega_N \approx \omega_d$ ✓

(xii) Two sound waves are $y = A \sin(\omega t - kx)$ and $y = A \cos(\omega t - kx)$. The phase difference between the two waves is:

- (a) $\pi/2$ ✓ (b) $\pi/4$ (c) π (d) 0

(xiii) 6. A ray of light passes from air into water. Striking the surface of the water with an angle of incidence 45° . Which of these quantities change as the light enters the water.

- i) Wavelength ii) frequency
iii) speed of propagation iv) direction of propagation
(a) i and ii only. (b) iii and iv only.
(c) i, iii, and iv only. ✓ (d) all of them.

(xiv) A communication channel consists of:

- (a) transmission line only (b) optical fibre only

- (c) free space only
 (xv) Dimension of ratio between angular momentum and momentum is:
 (a) L^{-1} ✓ (b) ML (c) L (d) L^{-2}
 (xvi) Work energy equation is called:
 (a) Law of conservation of mass
 (b) Law of conservation of momentum
 (c) Law of conservation of energy ✓
 (d) Law of conservation of angular momentum
 (xvii) The range of audible sound is:
 (a) 1 Hz to 10 Hz
 (b) 20 Hz to 20,000 Hz ✓
 (c) 21,000 Hz to 24,000 Hz
 (d) 25,000 Hz and onwards

SECTION 'B' (Short-Answer Questions) (32)

NOTE: Answer any NINE part questions from this section. questions carry equal marks. Draw diagrams where necessary.
 2 (i) Differentiate systematic error and random error

ANSWER:

RANDOM ERROR	SYSTEMATIC ERROR
It is the errors that fluctuate because of the uncertainty or unpredictability that is inherent in your measuring process or the differences in the quantity that you're trying to measure.	It occurs mostly because of apparatus' imperfection. They usually result from equipment that isn't correctly calibrated.
The magnitude of error keeps on changing in every reading.	The measured value will be very low or very high when compared to the true value.
This occurs on both sides.	This occurs only in one direction.

2 (ii) A car starts from rest and moves with a constant acceleration. In the 5th second of its motion, it covers a distance of 36 meters. Calculate

- (a) acceleration of the car
 (b) the total distance covered by the car during this time.

ANSWER:

DATA:

Initial Velocity = $V_i = 0$

time = $t = 5 \text{ sec}$

Distance during 5th sec = $S_5 = 36 \text{ m}$

Total Distance = $S = ?$

Acceleration = $a = ?$

SOLUTION:

Distance covered by the car in 4 seconds:

Distance covers in 4 sec = S_4

Total distance = $S = S_4 + S_5$

$$S = S_4 + 36\text{m}$$

$$S_4 = S - 36\text{m} \dots \dots (2)$$

Using 2nd equation of motion

$$\therefore S_4 = V_i + \frac{1}{2}at^2$$

$$S_4 = (0) + \frac{1}{2}a(4)^2$$

$$S_4 = 8a$$

Total Distance covered by the car in 5 seconds:

$$S = V_i + \frac{1}{2}at^2$$

$$S = (0) + \frac{1}{2}a(5)^2$$

$$S = 12.5a \dots \dots (2)$$

Substitute the value of S_4 and S in equation (1)

$$8a = 12.5a - 36$$

$$4.5a = 36$$

$$a = \frac{36}{4.5}$$

$$a = 8 \text{ m/s}^2$$

Substituting the value of 'a' in equation (2)

$$S = 12.5(8)$$

$$S = 100\text{m}$$

2 (iii) How does friction influence the motion of an object?

ANSWER: Friction is the force directed opposite to the direction of motion and its acting on an object is directly proportional to the area of contact between the object and the surface over which it executes motion. If the contact area is large and rough then more friction is produced, so the object moves slowly, similarly If contact area small and smooth then object moves fast.

2 (iv) A car accelerates uniformly from rest and reaches a speed of 22 m/s in 9s. If the diameter of a tire is 58 cm, find

(a) the number of revolutions the tire makes during this motion, assuming no slipping, and

(b) the final rotational speed of the tire in revolutions per second.

ANSWER:

DATA:

Initial Velocity = $V_i = \omega_i = 0$

Final Velocity = $V_f = V = 22 \text{ ms}^{-1}$

Time = $t = 9 \text{ sec}$

$$\text{Diameter of a tire} = d = 58\text{cm} = \frac{58}{100} = 0.58\text{m}$$

$$\text{Radius of a tire} = r = \frac{d}{2} = \frac{0.58}{2} = 0.29\text{m}$$

No. of Revolutions = ?

Final rotational speed = ?

SOLUTION:

For No. of Revolutions

$$\therefore V = r\omega$$

$$\omega_f = \frac{V}{r}$$

$$\omega_f = \frac{22}{0.29}$$

$$\omega_f = 75.862 \text{ rad/sec}$$

$$\omega_f = 75.862 \text{ rad/sec}$$

$$\therefore \omega_{avg.} = \frac{\omega_i + \omega_f}{2}$$

$$\omega_{avg.} = \frac{0 + 75.862}{2}$$

$$\omega_{avg.} = 37.93 \text{ rad/sec}$$

$$\theta = \omega t$$

$$\theta = (37.93)(9)$$

$$\theta = 341.379 \text{ rad}$$

$$\therefore \text{No of revolutions} = \frac{\theta}{2\pi}$$

$$\text{No of revolutions} = \frac{341.379}{2(3.142)}$$

$$\text{No of revolutions} = 54.33 \text{ rev}$$

For Rotational Speed:

$$\therefore V = r\omega$$

$$\omega_f = \frac{V}{r}$$

$$\omega_f = \frac{22}{0.29}$$

$$\omega_f = 75.862 \text{ rad/sec}$$

$$\omega_f = 75.862 \text{ rad/sec}$$

$$\omega_f = \frac{75.862}{2\pi}$$

$$\omega_f = 12 \text{ rev/sec}$$

2 (v) What is the work-energy theorem and how is it mathematically?

ANSWER:

WORK-ENERGY THEOREM

It states that "The total work done on the body is equal to the change in kinetic energy".

(The provided body is confined to move horizontally and no dissipating forces are operating).

MATHEMATICAL EXPRESSION:

Consider a body of mass m moving with initial velocity v_1 after travelling through displacement s its final velocity becomes v_2 under the effect of force F .

As we know that

$$2as = v_2^2 - v_1^2$$

$$a = \frac{v_2^2 - v_1^2}{2s}$$

Hence external force acting on the body is

$$F = ma$$

$$F = m \left(\frac{v_2^2 - v_1^2}{2s} \right)$$

Therefore work done on body by external force is

$$W = \vec{F} \cdot s \cos \theta$$

$$W = m \left(\frac{v_2^2 - v_1^2}{2s} \right) s \cos(0)$$

$$W = m \left(\frac{v_2^2 - v_1^2}{2} \right) (1)$$

$$W = \frac{mv_2^2 - mv_1^2}{2}$$

$$W = \frac{mv_2^2}{2} - \frac{mv_1^2}{2}$$

$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\therefore K.E. = \frac{1}{2}mv^2$$

$$W = K.E._2 - K.E._1$$

$$W = \Delta K.E$$

(i) The small piston of hydraulic press has an area of 10.0 cm^2 . If the applied force is 50.0 N , what must the area of the large piston to exert a pressing force of 4800 N ?

ANSWER: (See Question No. 2(vii), 2024 P.E. & P.M.)

(ii) State Bernoulli's principle.

ANSWER:

BERNOULLI'S PRINCIPLE

Bernoulli's principle states that "The velocity of a fluid is high, the pressure is low, and the velocity is low, the pressure is high".

OR

Bernoulli's principle states that "The total mechanical energy of the moving fluid comprising the gravitational potential energy of elevation, energy associated with the fluid pressure and the kinetic energy of motion, remains constant".

2 (viii) Two large parallel metal plates are 1.5 cm apart and have charges of equal magnitudes but opposite signs on their facing surfaces. Take the potential of the negative plate to be zero. If the potential halfway between the plates is then 5.0 V, what is the electric field in the region between the plates?

ANSWER:**DATA:**

Distance between Parallel plates = $D = 1.5 \text{ cm} = 0.015 \text{ m}$

Potential Difference of first plate = 5.0 volts

Potential Difference of half way of 2nd plate = 5.0 volts

Potential Difference between plates = $\Delta V = V_2 - V_1 = 5 - 0$
 $= 5 \text{ volts}$

Electric field = $E = ?$

SOLUTION:

$$E = \frac{\Delta V}{d}$$

Since the distance between the negative plate and the halfway point is the distance between the plates, i.e.

$$d = \frac{D}{2} = \frac{0.015}{2} = 0.0075 \text{ m}$$

$$E = \frac{5}{0.0075}$$

$$E = 666.67$$

$$E = 6.6667 \times 10^2 \approx 6.67 \times 10^2 \text{ V.m}^{-1}$$

2 (ix) State the factors on which the capacitance of a parallel plate capacitor depends.

ANSWER:**FACTORS ON WHICH THE CAPACITANCE OF A PARALLEL PLATE CAPACITOR DEPENDS**

The capacitance depends on the physical characteristics of the capacitor such as the area of the plates, the distance between them, and the properties of the dielectric material.

2 (x) The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.50Ω , what is the maximum current that can be drawn from the battery?

ANSWER: (See Question No. 2(x) Yousufi's Model Paper 1)

2 (xi) Discuss the factors that affect the period of a simple pendulum. Explain how the length of the pendulum, the acceleration due to gravity, mass and the amplitude of oscillation influence the period

ANSWER:

FACTORS THAT AFFECT THE PERIOD OF A SIMPLE PENDULUM:

The period of a simple pendulum, which is the time it takes for the pendulum to complete one full swing back and forth, is determined by a few key factors. The most significant of these is the length of the pendulum and other factor is the acceleration due to gravity.

EFFECT OF LENGTH

The period of a simple pendulum is directly proportional to the square root of the length. This means that if the length of the pendulum is increased, the period will also increase. Conversely, if the length is decreased, the period will decrease.

EFFECT OF ACCELERATION DUE TO GRAVITY

The acceleration due to gravity, g , also plays a crucial role in determining the period of a simple pendulum. The period is inversely proportional to the square root of g . Therefore, in a location where the acceleration due to gravity is higher, the period of the pendulum will be shorter, and vice versa. This is why a pendulum will swing slower at higher altitudes where gravity is slightly weaker.

EFFECT OF MASS AND THE AMPLITUDE

The time period of a simple pendulum is not affected by the mass and the amplitude of the pendulum.

(xii) A string 2.0 m long is held fixed at both ends. If a sharp blow is applied to the string at its centre, it takes 0.050 s for the pulse to travel to both ends of the string and return to the middle. What is the fundamental frequency of oscillation for this string?

ANSWER:

DATA:

Length of string = $L = 2.0\text{ m}$

time period = $t = 0.050\text{ s}$

fundamental frequency = $f_1 = ?$

SOLUTION:

$$\therefore v = \frac{L}{t}$$

$$v = \frac{2.0}{0.050}$$

$$v = 40 \text{ ms}^{-1}$$

$$\therefore f_1 = \frac{v}{\lambda_1}$$

$$f_1 = \frac{40}{\frac{2L}{20}}$$

$$f_1 = 10 \text{ Hz}$$

2 (xiii) Explain how the double-slit experiment provides evidence for wave nature of light.

ANSWER: The double-slit experiment provides evidence for the wave nature of light by the interference of light. In this experiment light comes from two different slits having specific path difference and interfere constructively or destructively. This interference supports the wave nature of light because particles do not interfere in this manner.

2 (xiv) Give 5 advantages of satellite communication.

ANSWER:

ADVANTAGES OF SATELLITE COMMUNICATION

- (i). Through satellite transmission, coverage over geographical areas is quite large mainly for sparsely populated areas.
- (ii). In satellite transmission High bandwidth and broadcast is possible.
- (iii). Wireless and mobile communication applications can be easily established by satellite communication independent of location.
- (iv). Security in satellite transmission is usually provided by the coding and decoding equipment.
- (v). High-quality components are used to work in robust conditions and links are also designed to work in extreme weather conditions.

SECTION "C" (Detailed Answer Questions)(3)

NOTE: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

3. (a) Explain Newton's Second Law of Motion and how it relates to mass, and acceleration. Provide an example to illustrate the concept.

ANSWER: According to Newton's Second law of motion, The acceleration of an object is directly proportional to the net force acting on it and is inversely proportional to its mass. Mathematically,

$$F \propto a$$

$$F = ma$$

The above relation shows that the acceleration of an object is directly proportional to the net force applied to it if the mass of object is constant.

constant. To understand this phenomenon, consider an object of mass "m" placed smooth and frictionless surface. When we applied force on it, it moves with certain velocity and acceleration will be produced. The change in velocity or acceleration will be directly proportional to the applied force. Similarly for constant acceleration applied force is directly proportional to the mass. For greater mass, more force is needed to produce acceleration, i.e.

$$F = ma$$

$$F \propto m$$

3. (b) Calculate gravitational potential energy at a certain height due to work against gravity.

ANSWER: We can calculate gravitational potential energy at a certain height due to work against gravity with the help of the following derivation.

DERIVATION

Consider a body of mass 'm' which is taken very slowly to a small height 'h' in the gravitational field such as that the acceleration of the body is zero. The work done in moving the body is given by:

$$\text{Work done} = F_{ex}h$$

$$W_{ex} = F_g h \cos 0$$

Where ' F_{ex} ' is the external force applied on the body. Since the external force applied on the body and the displacement are along the same direction, therefore work done by external force ' W_{ex} ' is given by:

$$W_{ex} = F_{ex}h \cos 0$$

$$W_{ex} = F_g h (1)$$

$$W_{ex} = F_g h$$

As the acceleration of the body is zero therefore magnitude of external force is equal to that of the force of gravity i.e.

$$F_{ex} = mg$$

Therefore

$$W_{ex} = mgh \text{ --- (i)}$$

Work done ' W_g ' by the gravitational force ' F_g ' is given by

$$W_g = \vec{F}_g \cdot \vec{h} \cos \theta$$

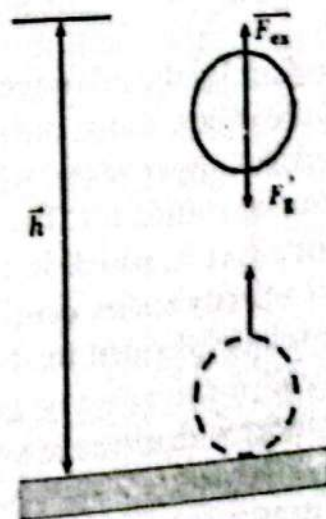
The angle between $\vec{F}_g$ and $\vec{h}$ is 180° , so

$$W_g = \vec{F}_g \cdot \vec{h} \cos 180^\circ$$

$$W_g = \vec{F}_g \cdot \vec{h} (-1)$$

$$W_g = -\vec{F}_g \cdot \vec{h}$$

$$\because F_g = mg$$



$$W_g = -mg \cdot \vec{h}$$

$$-W_g = mg \cdot \vec{h} \quad \text{--- (ii)}$$

On comparing equation (i) and (ii), we get

$$W_{ex} = -W_g$$

This work done on the body by an external force against the gravitational force is stored in the form of potential energy and is known as gravitational potential energy represented by U_g

Therefore

$$U_g = W_{ex}$$

$$\text{But } W_{ex} = -W_g$$

$$U_g = -W_g$$

$$U_g = mg \cdot \vec{h}$$

Hence by the above expression, we can calculate gravitational potential energy at a certain height due to work against gravity.

4. (a) Discuss viscous force in fluids.

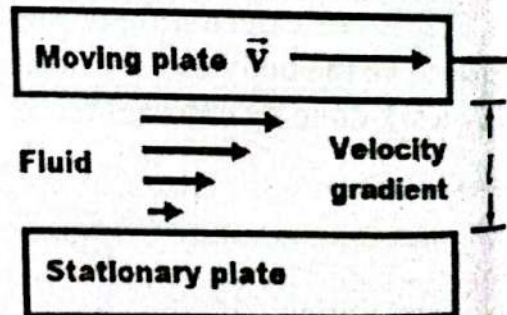
ANSWER:

VISCOUS FORCE IN FLUIDS

Viscous force is an opposition between layers of a fluid. Viscous forces in fluid are proportional to the rate of change of velocity of the fluid's layers. The viscosity of a fluid serves as the proportionality constant.

Suppose a viscous fluid between two plates in which one plate is stationary and other is moveable as shown in figure.

The fluid is directly in contact with each plate and is held to



Surface by the adhesive force between the molecules of the liquid and the plates. Thus, the upper surface of the fluid moves with the same velocity v as the upper plate, whereas the fluid in contact with the stationary plate remains stationary. The stationary layer of fluid retards the flow of the layer just above it, which in turn retards the flow of next layer, and so on. Thus, the velocity varies continuously from 0 to v through the fluid. The rate of change in velocity divided by the distance over which change is made - equal to the velocity gradient which is defined as 'the rate of change of velocity with distance normal to the direction of flow of the layers of the fluid with respect to object passing through the fluid'.

Viscosity is a fluid's resistance to flow. Fluids resist the relative motion of the immersed objects through them as well as to the motion of layers having differing velocities within them.

For a given fluid, it is required force F which is proportional to the area of fluid in contact with each plate 'A', and to the speed, ' v ', and is inversely proportional to the separation ' l ', between the plates as shown in figure

$$F \propto v \frac{A}{l}$$

For more viscous fluid, the greater force is the required. Hence, the proportionality constant for this equation is defined as the coefficient of viscosity η which is intrinsic property of the fluid.

$$F = \eta A \frac{v}{l}$$

The temperature has a strong effect on the viscosity of liquids, such as the viscosity of motor oil decreases rapidly as temperature increases, while for gases it increases.

4. (b) Define an electric dipole. Derive formula for the electric field due to an electric dipole at a point "P" placed on its axial line.

ANSWER: (See Question No. 3(b), 2024 P.E. & P.M.)

5. (a) Discuss the factors that affect the sharpness of resonance in an oscillatory system. Explain how damping and quality factor influence the width and peak of the resonance curve.

ANSWER:

FACTORS THAT AFFECT THE SHARPNESS OF RESONANCE

The sharpness of resonance depends mainly on two factors: amplitude and damping.

DAMPING AND QUALITY FACTORS INFLUENCE THE WIDTH AND PEAK OF THE RESONANCE CURVE:

The Q-factor quantifies the sharpness of resonance. It signifies the reduction of the oscillation's amplitude over time, which corresponds to the decay of energy in an oscillating system. It is approximately defined as the number of free oscillations the oscillator undergoes before its amplitude decays to zero. In the case of light damping, the Q-factor will be large, whereas it will be small for significant damping. Mathematically, the Q-factor is the ratio of energy stored to energy lost per oscillation, and it is a dimensionless quantity.

$$Q = \frac{E_{\text{stored}}}{E_{\text{lost}}}$$

5. (b) Define the Doppler Effect and Derive the Doppler Effect equation for sound waves in terms of the relative velocity between the source, observer, and the speed of sound

ANSWER:

DOPPLER EFFECT

The Doppler Effect is a phenomenon in physics that describes the perceived change in frequency of a wave (such as sound or light) when the source of

the wave or the observer is in relative motion. In the case of sound the Doppler Effect occurs when there is relative motion between the source of the sound and the observer.

DOPPLER EFFECT EQUATION

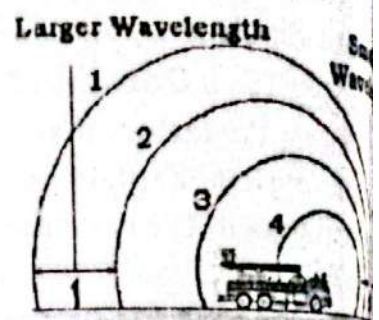
A source emits a sound wave at frequency f , with velocity v and wavelength λ which means that wave crests (regions of maximum amplitude), indicated by circles in leave the source spaced by a time interval

$$T = \frac{1}{f}$$

MOVING SOURCE:

SOURCE IS MOVING TOWARDS THE OBSERVER AT REST:

If the source is moving at velocity v_s toward a stationary observer on the right, the first figure shows that the wavelength, the distance between the crests is smaller to the right. The second Figure shows that at the instant that crest 4 is emitted, crest 3 has travelled outward



a distance vT_s from point 3. During the same time interval, the source has advanced a distance v_sT_s . The wavelength λ as measured by the observer on the right is the distance between crests 4 and 3:

$$\lambda = vT_s - v_sT_s$$

The frequency at which the crests arrive at the observer is the observed wave frequency f' .

The observed period T_o between the arrival of two crests is the time

it takes sound to travel a distance $(v - v_s)T_s$:

$$T_o = \frac{(v - v_s)T_s}{v}$$

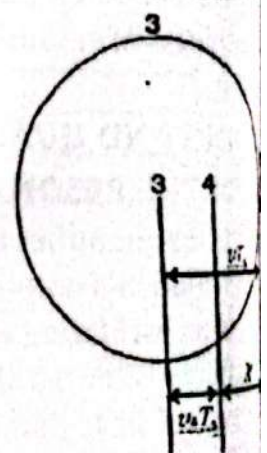
The observed frequency is

$$f' = \frac{1}{T_o} = \frac{1}{\frac{(v - v_s)T_s}{v}}$$

$$f' = \frac{v}{(v - v_s)T_s} = \frac{v}{(v - v_s)} \frac{1}{T_s}$$

$$f' = \frac{v}{(v - v_s)} f_s$$

SOURCE IS MOVING AWAY FROM THE OBSERVER AT REST:



If the source is moving at velocity v_s away from the stationary observer on the left, then according to figure, the wavelength, the distance between the crests is larger to the left. The wavelength λ' as measured by the observer on the left (in above Figure) is the distance between crests 3 and 4:

$$\lambda' = vT_s + v_s T_s$$

The frequency at which the crests arrive at the observer is the observed wave frequency f'' .

The observed period T_o between the arrival of two crests is the time it takes sound to travel a distance $(v + v_s) T_s$:

$$T_o = \frac{(v + v_s) T_s}{v}$$

The observed frequency is

$$f'' = \frac{1}{T_o} = \frac{1}{\frac{(v + v_s) T_s}{v}}$$

$$f'' = \frac{v}{(v + v_s) T_s}$$

$$f'' = \frac{v}{(v + v_s) T_s} \cdot 1$$

$$f'' = \frac{v}{(v + v_s)} f_s$$

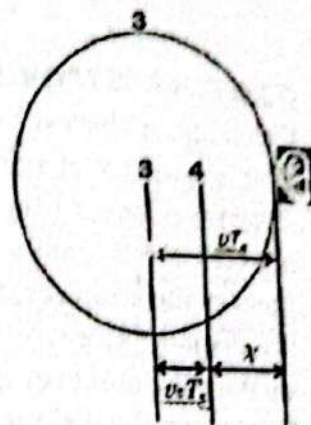
Hence; the observed frequency by the stationary listener shall be less than the source frequency, given as

MOVING OBSERVER:

OBSERVER IS MOVING TOWARDS THE SOURCE AT REST:

A stationary source emits a sound wave at frequency f_s and wavelength $\lambda_s = \frac{v}{f_s}$, where v is the speed of sound. An observer moving towards the

stationary source with speed v_o , would observe a shorter time interval between crests. Just as crest 1 reaches the observer, the next crest 2 is at distance λ_s ahead. Crest 2 catches up with the observer with time T' earlier than the time T_s , when the distance vT' the wave crest 2 travels toward the observer is equal to the wavelength of sound minus the distance the observer travels towards the wave crest 2 as shown in figure



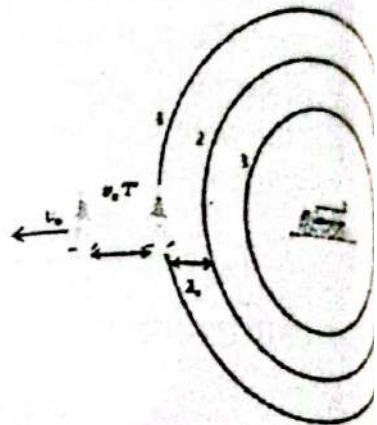
$$\begin{aligned} vT' &= \lambda_s - v_o T' \\ vT' + v_o T' &= \lambda_s \\ \lambda_s &= (v + v_o)T' \end{aligned}$$

Given that

$$\begin{aligned} T' &= \frac{1}{f'''} \quad \text{and} \quad \lambda_s = \frac{v}{f_s} \\ \frac{v}{f_s} &= (v + v_o) \frac{1}{f'''} \\ f''' &= \frac{(v + v_o)}{v} f_s \end{aligned}$$

OBSERVER IS MOVING AWAY FROM THE SOURCE AT REST.

Consider an observer moving away from the source at velocity v_o . He observes a longer time interval between crests. Just as crest 1 reaches the observer, the next crest 2 is a distance λ_s away. Crest 2 catches up with the observer at a time T'' later when the distance the wave crest



travels toward the observer is equal to the distance the observer travels away from the wave crest plus the wavelength as shown in figure

$$\begin{aligned} vT'' &= v_o T'' + \lambda_s \\ vT'' - v_o T'' &= \lambda_s \\ \lambda_s &= (v - v_o)T'' \end{aligned}$$

Given that

$$\begin{aligned} T'' &= \frac{1}{f''''} \quad \text{and} \quad \lambda_s = \frac{v}{f_s} \\ \frac{v}{f_s} &= (v - v_o) \frac{1}{f''''} \\ f'''' &= \frac{(v - v_o)}{v} f_s \end{aligned}$$

An observer moving away from the source detects a frequency lower than the f_s .

MOTION OF BOTH SOURCE AND OBSERVER:

Since the relative change in the frequency (pitch) of sound with respect to the listener is dependent upon the motion of source of sound or listener.

SOURCE AND LISTENER BOTH MOVE TOWARDS EACH OTHER

If the source and listener both move towards each other then we get the two Doppler's shifts. First consider the relative change in frequency.

with respect to stationary listener as the source is approaching towards the listener. Let f_L be the frequency detected by the listener.

$$f_L = \left(\frac{v}{v - v_s} \right) f_s$$

If at some moment the listener starts moving towards the approaching source with velocity v_o , then the detected frequency f_o shall be

$$f_o = \left(\frac{v + v_o}{v} \right) f_L$$

Substituting the value of f_L yields

$$f_o = \left(\frac{v + v_o}{v} \right) \left\{ \left(\frac{v}{v - v_s} \right) f_s \right\}$$

$$f_o = \left(\frac{v + v_o}{v - v_s} \right) f_s$$

SOURCE AND LISTENER BOTH MOVES AWAY FROM EACH OTHER

If the source and listener both moves away from each other then we combine the two Doppler's shifts. First consider the relative change in frequency with respect to stationary listener as the source is moving away from the listener. Let f_L' be the frequency detected by the listener.

$$f_L' = \left(\frac{v}{v + v_s} \right) f_s$$

If at some moment the listener starts moving away from the source with velocity v_o , then the detected frequency f_o' shall be

$$f_o' = \left(\frac{v - v_o}{v} \right) f_L'$$

Substituting the value of f_L yields

$$f_o' = \left(\frac{v - v_o}{v} \right) \left\{ \left(\frac{v}{v + v_s} \right) f_s \right\}$$

$$f_o' = \left(\frac{v - v_o}{v + v_s} \right) f_s$$

YOUSUIFS MODEL QUESTION PAPER 4

PHYSICS (THEORY)

CLASS XI (SCIENCE)

Total Time: 3 Hours

Total Marks: 85

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

NOTE: This Section consists of 17 MCQs and all are to be answered. Each MCQ carry 1 marks.

Choose the correct answer from the given options:

[MLT⁻²] is dimensional formula of:

- (a) strain (b) force ✓ (c) displacement (d) pressure
- (ii) If the dot product of two non-zero vectors vanishes; the vectors must be:
- (a) in the same direction (b) opposite direction to each other ✓
(c) perpendicular to each other ✓ (d) zero
- (iii) The kinetic energy of body of mass 2 kg and momentum of 2 kg m/s is:
- (a) 1 J ✓ (b) 2 J (c) 3 J (d) 4 J
- (iv) One radian is about:
- (a) 25° (b) 37° (c) 45° (d) 57° ✓
- (v) The escape velocity of a planet depends on which of the following factors?
- (a) The mass of the planet only
(b) The radius of the planet only
(c) Both the mass and the radius of the planet ✓
(d) The density of the planet
- (vi) A completely submerged object always displaces its own:
- (a) weight of fluid (b) volume of a fluid ✓
(c) density of a fluid (d) Area of fluid
- (vii) Bernoulli's principle states that for horizontal flow of a fluid through a tube, the sum of the pressure and energy of motion per unit volume:
- (a) increasing with time (b) decreasing with time
(c) constant ✓ (d) varying with time
- (viii) The minimum charge on an object cannot be less than:
- (a) $1.6 \times 10^{-19}\text{ C}$ ✓ (b) $3.2 \times 10^{-19}\text{ C}$
(c) 9.1×10^9 (d) No definite value exists
- (ix) The time constant of a series RC circuit consisting of a capacitor in series with a 100Ω resistor is:
- (a) 0.1 s (b) 0.1 ms (c) 0.01 s ✓ (d) 0.01 ms
- (x) Kirchhoff's laws are useful in determining:
- (a) current flowing in a circuit ✓
(b) emf and voltage drops in a circuit
(c) power in a circuit (d) only emf in a circuit
- (xi) Two simple pendulums A and B with same lengths, and equal amplitude of vibrations, but the mass of A is twice the mass of B. Their periods are T_A and T_B and energies are E_A and E_B respectively. The correct statement is:
- (a) $T_A = T_B$ and $E_A > E_B$ ✓
(b) $T_A < T_B$ and $E_A > E_B$
(c) $T_A > T_B$ and $E_A < E_B$ (d) $T_A = T_B$ and $E_A < E_B$
- (xii) The speed of sound in air at STP is 332 m/s . If the air pressure becomes double at the same temperature, the speed of sound becomes:

- 1382 m/s ✓
 1382 m/s ✓
 (b) 664 m/s
 (d) 166 m/s
 Plane polarized light is passed through a Polaroid. On viewing through the Polaroid we find that when Polaroid is given one complete rotation about the direction of light
 The intensity of light gradually decreases to zero and remains at zero.
 The intensity of light gradually increases to maximum and remains maximum.
 There is no change in the intensity of light.
 The intensity of light varies such that it is twice maximum and twice
 What is the frequency range of signals that can be transmitted in case of twisted pair of wires?
 10 MHz to 15 MHz.
 100 Hz to 5 MHz. ✓
 Dimension of pressure is:
 (b) 5 MHz to 10 MHz.
 (d) 10 Hz to 100 Hz.
 (b) $ML^{-2}T^{-3}$
 (d) MLT^{-1}
 A weight lifter consumes 500 J of energy to lift a load in 2 seconds, the power consumed is:
 125 watt (b) 500 watt (c) 250 watt ✓ (d) 1000 watt
 The motion of number of lines per cm of a diffraction grating are 100, its grating element is:
 (b) $4 \times 10^{-2} cm$
 (d) $4 \times 10^5 cm$
 (b) $2.5 \times 10^{-4} cm$ ✓
 (d) $2.5 \times 10^{-6} cm$

SECTION 'B' (Short-Answer Questions) (32)

NOTE: Answer any NINE part questions from this section. All questions carry equal marks. Draw diagrams where necessary.

- (i) A rectangular holding tank 25.0 m in length and 15.0 m in width is used to store water for short period of time in an industrial plant. If 2980 M^3 water is pumped into the tank. What is the depth of the water?

ANSWER:

DATA:

- length of tank = $l = 25.0 m$
 width of tank = $w = 15.0 m$
 Volume of water (tank) = $2980 m^3$
 depth of tank = $d = ?$

SOLUTION:

$$\therefore V = l \times w \times d$$

$$\frac{v}{l \times w} = d$$

$$d = \frac{25 \times 15}{2980}$$

$$d = \frac{375}{2980}$$

$$d = 7.9466 \approx 7.95m$$

2 (ii) If air resistance is taken into account in case of projectile motion what parameters of projectile are influenced?

ANSWER: When air resistance is present, the range of the projectile maximum height will be reduced, but the total time of flight will increase. Air resistance also affects the angle of projection.

2 (iii) A machine gun fires 10 bullets per second into a target. Each bullet weighs 20 gram and had a speed of 1500 m/s. Find the necessary force to hold the gun in position?

ANSWER:

DATA:

No. of Bullets = 10

Mass of each bullet = $m = 20g = \frac{20}{1000} = 0.02kg$

Mass of 10 bullet = $M = 0.02 \times 10 = 0.2kg$

Speed of bullets = $1500ms^{-1}$

Time = $t = 1 \text{ second}$

Force to hold the gun = $F = ?$

SOLUTION:

$$\therefore \vec{F} = m\vec{a}$$

$$F = M \left(\frac{d\vec{v}}{dt} \right)$$

$$F = (0.2) \left(\frac{1500}{1} \right)$$

$$F = 300 N$$

2 (iv) When the moment of inertia of a rotating body is halved, then what will be the effect on angular velocity?

ANSWER: We know that the moment of inertia of a rotating body is

$$I = \frac{L}{\omega}$$

Dividing both sides by "2" so that moment of inertia of rotating body becomes half

$$\frac{I}{2} = \frac{L}{2\omega}$$

The above relation shows that "when the moment of inertia of a rotating body is halved then its angular velocity will be doubled".

Q. A motor pumps the water at the rate 500 gram/minute to the height of 120 m. If the motor is 50% efficient then how much input electric power is needed?

ANSWER:

DATA:

Rate = 500 gram/minute

Height = $h = 120\text{m}$

Efficient = 50%

Acceleration due to gravity = $g = 10\text{ ms}^{-2}$

Input Electric power = ?

SOLUTION:

$$\begin{aligned}\therefore \text{Rate} &= \frac{m}{t} \\ \frac{m}{t} &= \frac{500\text{g}}{1\text{ min}} \\ \frac{m}{t} &= \frac{500\text{g}}{60} \times \frac{1}{1000} \\ \frac{m}{t} &= \frac{1}{120} \text{ kg/sec}\end{aligned}$$

$$120 \times m = t$$

$$t = 120 \times m$$

$$P = \frac{mgh}{t}$$

$$P = \frac{mgh}{120 \times m}$$

$$P = \frac{gh}{120}$$

$$P = \frac{10 \times 120}{120}$$

$$P = 10 \text{ watt}$$

$$\therefore \text{Efficiency} = \frac{\text{Output}}{\text{Input}} \times 100$$

$$\text{Input} = \frac{\text{Output}}{\text{Efficiency}} \times 100$$

$$\text{Input} = \frac{10}{50} \times 100$$

$$\text{Input} = 20 \text{ watt}$$

2. (vi) Define surface tension and give its any two applications.

ANSWER:

SURFACE TENSION

Surface tension is defined as

"The force per unit length acting on either side of the imaginary line on the liquid surface at rest. The direction of force is tangential to the surface and perpendicular to the line".

$$\text{Surface Tension} = \frac{F}{L}$$

Its SI unit is N/m or Nm^{-1} .

APPLICATIONS

1. When a pin is placed gently over the surface of water, it floats. The depression of the surface which behaves like a stretched membrane. If a needle is pressed with a finger, the layer of surface tension breaks and the needle sink in the water.
2. Some insects (with long legs) can walk over the water surface. Surface tension of water supports the weight of insect. Actually insect sinks slightly into the fluid so its weight becomes effective weight of that insect, whose weight is less than the buoyant force.
- 2 (vii) A liquid flows through a pipe with a diameter of 0.50 m at a speed of 4.20 m/s . What is the rate of flow in L/min ?

ANSWER:

DATA:

Diameter of pipe = $D = 0.50\text{ m}$

Radius of pipe = $r = \frac{D}{2} = \frac{0.50}{2} = 0.25\text{ m}$

Cross - section area of liquied flow = $A = \pi r^2 = \pi(0.25)^2$
 $= (0.0625)\pi = 0.196\text{ m}^2$

Speed of liquid = $v = 4.2\text{ ms}^{-1}$

Rate of flow = $Q = ?$

SOLUTION:

$$\therefore Q = Av$$

$$Q = (0.196)(4.2)$$

$$Q = 0.825\text{ m}^3\text{s}^{-1}$$

$$Q = 0.825 \times 1000 \times 60$$

$$Q = 49480\text{ L.min}^{-1}$$

- 2 (viii) What is electric dipole and electric dipole moment?

ANSWER:

ELECTRIC DIPOLE:

An electric dipole is a simple system in electromagnetism consisting of two opposite electric charges of equal magnitude, separated by a small distance d . The charges create an electric field that has a distinct pattern, with field lines oriented along the axis of the dipole.

ELECTRIC DIPOLE MOMENT

The Electric dipole moment $\vec{p}$ is a vector quantity that represents the strength and direction of the electric dipole. It is defined as "the product of the charge magnitude (q) and the distance between the charges d ".

$$\vec{p} = q \times d$$

Electric dipoles play a fundamental role in understanding the behavior of polar molecules, interactions in electric fields, and various electrical phenomena in physics and engineering.

(ix) A $2.0 \mu F$ capacitor and a $4.0 \mu F$ capacitor are connected in parallel across a $300 V$ potential difference. Calculate the total energy stored in the capacitors.

ANSWER:

DATA:

Capacitance of 1st capacitor = $C_1 = 2 \mu F$

Capacitance of 2nd capacitor = $C_2 = 4 \mu F$

Potential difference = $V = 300 V$

Energy = $U = ?$

SOLUTION:

Since both capacitors are connected parallel, therefore

$$\therefore C = C_1 + C_2$$

$$C = 2 \mu F + 4 \mu F$$

$$C = 6 \mu F$$

$$C = 6 \times 10^{-6} F$$

$$\therefore U = \frac{1}{2} CV^2$$

$$U = \frac{1}{2} (6 \times 10^{-6}) (300)^2$$

$$U = (3 \times 10^{-6}) (90000)$$

$$U = 0.27 J$$

2 (x) Why is a potentiometer preferred over a voltmeter for determining the emf of a cell?

ANSWER: A potentiometer is preferred over voltmeter to measure emf of a cell because it has high sensitivity as compared to a voltmeter meter and it measures true emf of the cell. Also, the potentiometer does not draw any current while measuring emf whereas the voltmeter draws current to measure emf.

2 (xi) An ideal spring with a spring constant of $15 N/m$ is suspended vertically. A body of mass $0.60 kg$ is attached to the upstretched spring and released.

(a) What is the extension of the spring when the speed is a maximum?
(b) What is the maximum speed?

ANSWER:

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DATA:

spring constant = $k = 15 \text{ Nm}^{-1}$

Mass of body = $m = 0.60 \text{ kg}$

Acceleration due to gravity = 9.8 ms^{-2}

extension of the spring = $y = ?$

maximum speed = $v_o = ?$

SOLUTION:

When an object executes SHM, The speed of the object is maximum when it passes from its mean position therefore

$$\because F_{\text{net}} = ky - mg = 0$$

$$ky - mg = 0$$

$$ky = mg$$

$$y = \frac{mg}{k}$$

$$y = \frac{(0.60)(9.8)}{15}$$

$$y = \frac{5.88}{15}$$

$$y = 0.392 \text{ m}$$

For Maximum speed

$$\because \omega = \frac{2\pi}{T}$$

$$\omega = \frac{2\pi}{2\pi \sqrt{\frac{m}{k}}}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{15}{0.60}}$$

$$\omega = \sqrt{25}$$

$$\omega = 5 \text{ ms}^{-1}$$

$$\because v_o = \omega y$$

$$v_o = (5)(0.392)$$

$$v_o = 1.96 \approx 2 \text{ ms}^{-1}$$

2 (xii) Why is it possible to understand the words spoken by two people at the same time?

ANSWER: It is possible because all people have different pitches and quality of sound. Due to this difference we can understand the words spoken by two different people at the same time.

(xiii) Newton's rings are formed by the light of 400 nm wavelength. Determine the change in air film thickness between the third and sixth bright fringe. If the radius of curvature of the curved surface is 5.0 m what is the radius of third bright fringe?

ANSWER:

DATA:
Wavelength = $\lambda = 400 \text{ nm} = \frac{400}{10^9} = 4.0 \times 10^{-7} \text{ m}$

Radius of Curvature = $R = 5.0 \text{ m}$

No of ring = $n = 3, 6$

Radius of third bright fringe = $r_3 = ?$

Film thickness between the third and sixth bright fringe = $\Delta t = ?$

SOLUTION:

$$\therefore 2t_N = \frac{(2N - 1)\lambda}{2}$$

$$2t_3 = \frac{\{2(3) - 1\}(400)}{2}$$

$$t_3 = \frac{(6 - 1)(400)}{4}$$

$$t_3 = \frac{(5)(400)}{4}$$

$$t_3 = 500 \text{ nm}$$

$$2t_6 = \frac{\{2(6) - 1\}(400)}{2}$$

$$t_6 = \frac{(12 - 1)(400)}{4}$$

$$t_6 = \frac{(11)(400)}{4}$$

$$t_6 = 1100 \text{ nm}$$

$$\therefore \Delta t = t_6 - t_3$$

$$\Delta t = 1100 - 500$$

$$\Delta t = 600 \text{ nm}$$

$$\therefore r_N = \sqrt{R \left(N - \frac{1}{2} \right) \lambda}$$

$$r_3 = \sqrt{(5) \left(3 - \frac{1}{2} \right) (4.0 \times 10^{-7})}$$

$$r_3 = \sqrt{(5) \left(\frac{5}{2} \right) (4.0 \times 10^{-7})}$$

$$r_3 = \sqrt{5.0 \times 10^{-6}}$$

$$r_3 = 2.236 \times 10^{-3} \text{ m}$$

$$r_3 = 2.236 \times 10^{-3} \times 10^{-3}$$

$$r_3 = 2.236 \text{ mm}$$

2 (xiv) What are limitations of amplitude modulation?

ANSWER: (See Question No. 2 (xiv) Yousufi's Model Paper 2)

SECTION "C" (Detailed Answer Questions)(32)

NOTE: Attempt any TWO questions from this section. questions carry equal marks. Draw diagrams where necessary.

3. (a) Discuss graphs and its various types by supported an example?

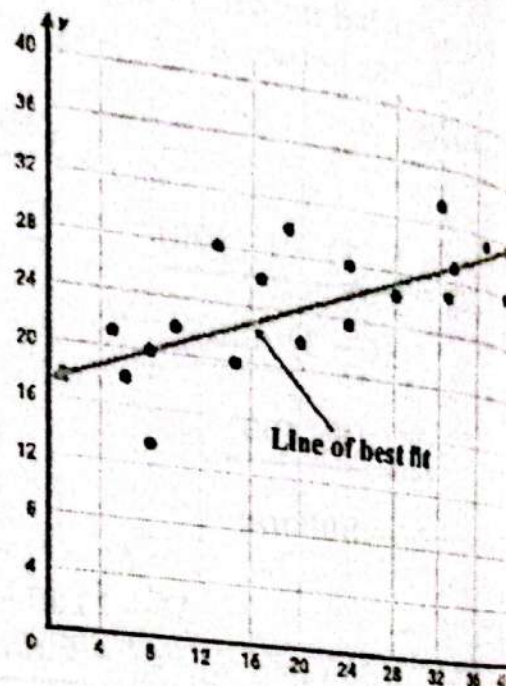
ANSWER:

GRAPH:

Graphs are visuals that show relationships between; intended to display data in a way that is easy to understand and remember. Graphs are used to demonstrate trends, patterns and relationships between sets of data. Graphs may be preferable to display certain types of data.

BEST FIT LINE GRAPH:

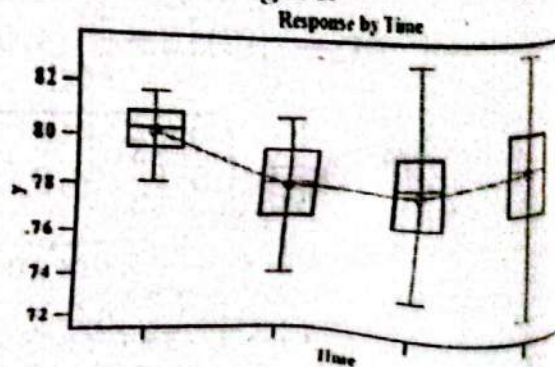
A best fit line graph is a type of graph used to visualize the relationship between two variables and is used to show the general trend in the data. The line of best fit is a straight line that is drawn in a way that it best represents the underlying pattern in the data.



ERROR BAR:

Error bars are graphical representations of the uncertainty or variability in a set of data points. They are often used in scientific plots to indicate the precision of the data being plotted as shown in figure.

Error bars are usually plotted as vertical or horizontal lines extending from the data points on a graph. The length of the error bars indicates the degree of uncertainty or variability in the data. Shorter error bars indicate



smaller precision, while longer error bars indicate less precision. Error bars are useful in indicating the reliability and accuracy of the data, and they allow the reader to assess the significance of the results. In scientific research, error bars are often used to determine the statistical significance of differences between groups or to compare the results of different experiments.

(b) State and explain the law of conservation of angular momentum. Give two examples to illustrate it.

ANSWER:

STATEMENT

The law of conservation of angular momentum states that "when no resultant external torque acts on a body, its angular momentum remains constant".

EXPLANATION:

Consider a particle in a uniform circular motion due to a force acting on the particle that is centripetal force required to deflect the particle to keep it in circular path. The centripetal force always points towards the center of the circle so it produces no torque about an origin at the center of the circle.

The law of conservation of angular momentum then ensures that the angular momentum of the particle is constant. The angular momentum is constant in magnitude ($mr\omega^2$ remains fixed) and constant in direction (the motion is confined to the plane of rotation).

EXAMPLES

The spinning of the Earth on its axis at the same rate for billions of years. The speed of the earth is more near the sun and comparatively less when it is away from the sun, in an elliptical orbit. Due to this reason, the angular momentum of Earth is conserved.

EXAMPLES

In ice skating, an ice-skater changes his velocity by holding his arms outwards or pulling them inwards. When he pulls his hands in, his inertia decreases. Since there is no external force acting on him, angular momentum is conserved as his velocity magnitude increases.

4. (a) Discuss surface tension with at least three experiments.

ANSWER:

SURFACE TENSION

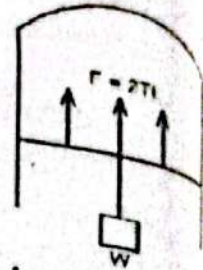
"The force per unit length acting on either side of the imaginary line drawn on the liquid surface at rest. The direction of force is tangential to the surface and perpendicular to the line".

$$\text{Surface Tension} = \frac{F}{L}$$

Its SI unit is N/m or $\text{N} \cdot \text{m}^{-1}$.

It is the ability of the surface of a liquid to act like a thin flexible film. surface of the water to support a needle, the water's surface acts like a flexible film

Consider a U-shaped apparatus which encloses a thin film of liquid (Liquid soap film). Because of surface tension, a force F is required to pull the movable wire and this increase the surface area of the liquid. The liquid contained by the wire apparatus is thin film having both top and



bottom surface. Hence the total length of the surface being increased is $2l$ and the surface tension is

$$T = \frac{F}{2L}$$

A delicate apparatus of this type can be used to measure the surface tension of various liquids.

EXPERIMENT 1:

A drop of Olive oil is dropped through a pipette, gently inside a mixture of alcohol and water, which forms a perfectly spherical shape. The density of the mixture is the same as of olive oil. Due to surface tension, the tendency of a liquid surface is to keep its area minimum. For a given volume, the surface area of a sphere is minimum. Due to this reason, the raindrop is spherical shape.

EXPERIMENT 2:

A glass tube of the small bore is dipped into water; it rises up the tube a few centimeters. As tube becomes narrower, there is more rise up. Adhesive force between water and glass exceeds cohesion force between water molecules, the meniscus causes up and surface tension.

EXPERIMENT 3:

A mercury is dropped over glass surface, the globules are spherical in shape because of surface tension.

4. (b) Describe the factors affecting the capacitance of a parallel plate capacitor?

ANSWER:

FACTORS ON WHICH THE CAPACITANCE OF A PARALLEL PLATE CAPACITOR DEPENDS

The capacitance depends on the physical characteristics of the capacitor such as the area of the plates, the distance between them, and the properties of the dielectric material.

EFFECT OF AREA OF THE PLATES:

The capacitance of a parallel plate condenser is directly proportional to the area of the plates. This means that as the area of the plates increases, the capacitance also increases.

capacitance also increases and vice versa. The formula shows that the capacitance is inversely proportional to the distance between the plates and the permittivity of the medium between the plates.

EFFECT OF DISTANCE BETWEEN THE PLATES:

The capacitance of a parallel plate condenser is inversely proportional to the distance between the plates. This means that as the distance between the plates increases, the capacitance decreases and vice versa. The formula shows that the capacitance is directly proportional to the area of the plates and the permittivity of the medium between the plates.

EFFECT OF MEDIUM BETWEEN THE PLATES:

The capacitance of a parallel plate condenser is directly proportional to the permittivity of the medium between the plates. This means that as the permittivity of the medium increases, the capacitance also increases and vice versa. The formula shows that the capacitance is inversely proportional to the distance between the plates and the area of the plates.

5. (a) Define thermoelectricity. Explain the working of a Fire Alarm system using thermistor.

ANSWER:

THERMOELECTRICITY

The electricity can be generated from heat energy and thermoelectricity is the electricity generated from heat energy. In thermoelectricity, the conversion of heat energy into electrical energy takes place with the help of a thermocouple.

WORKING OF A FIRE ALARM SYSTEM

A thermistor is a variable resistor whose resistance changes with temperature. Its temperature detection can be used in fire alarms for the detection of fires. Fire Alarm Circuit is a simple circuit that detects the fire and activates the Siren Sound or Buzzer. In this fire alarm circuit, the resistance of the thermistor is approximately kilohms at normal temperature. During fire, the resistance reduces to a few ohms as the temperature increases which switches ON the transistor. Once the transistor is turned ON, the current from Vcc starts to flow via buzzer which produces a beep sound. For unidirectional conduction a Diode is used and the use of capacitor removes sudden transients from the thermistor.

5. (b) What is modulation? Why it is needed for the transmission of signals?

ANSWER:

MODULATION

Modulation is the process of transmitting a message signal with a carrier signal to cover longer distances while maintaining message quality. It uses high-frequency carrier waves for energy and to overcome obstructions during transmission. Antennas translate audio signals to higher frequencies for effective operation. Modulation varies the carrier wave's characteristic

with time. Both analog and digital information can be encoded in modulation, enabling efficient communication systems like cell phones to convert voice into electrical signals for transmission and reception. Mathematically, a carrier wave is represented by a sinusoidal waveform

$$V_c(t) = A \sin(2\pi f_c t + \phi)$$

Where

A - represents the amplitude of the carrier wave

f_c - represents the frequency of the carrier wave

ϕ - represents the phase angle

NEED OF MODULATION

The baseband signals transmitted by the sender are not capable of direct transmission. The strength of the message signal should be increased so that it can travel longer distances. This is where modulation is essential. The most vital need of modulation is to enhance the strength of the signal without affecting the parameters of the carrier signal.

An audio signal usually has a frequency range (20 Hz to 20 kHz), if it is directly transmitted then the ratio of highest to the lowest frequency becomes $(20 \text{ kHz} / 20 \text{ Hz}) = 1000$. But if this audio signal is modulated on a carrier signal of frequency 1000 kHz then the ratio of highest to the lowest frequency becomes:

$$(1000 \text{ kHz} + 20 \text{ kHz}) / (1000 \text{ kHz} - 20 \text{ kHz}) \cong 1.2$$

Hence, we need modulation to convert a wideband signal into a narrow band signal.

It is practically not possible to distinguish between the different audio signals when transmitted simultaneously through a single antenna as all of them are in the same spectral range. Hence, each of these signals is translated to a different frequency range before transmission which makes it quite easier to receive them and distinguish each of them from one another at the receiver's end.